



CORA

A Framework for Quantitative Reliability Assessment of Subsurface Containment



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CORA
A Framework for Quantitative Reliability Assessment of Subsurface Containment
by
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Executive summary

The technical workflow and supporting tools are described to perform a quantitative assessment of containment risks associated with the injection and storage of CO₂ or H₂S in geological formations. The workflow aligns with the Opportunity Realisation Process (ORP) and supports the use of TESLA, the tool for risk management in storage projects. The quantitative assessment is focussed to geomechanical aspects of CO₂ storage.

The framework for Containment Risk Assessment tool CORA is described by means of its five modules that perform clearly defined tasks and can be adjusted to the project needs. The system reliability module includes an assessment tree, which specifies the key leakage pathways, the failure modes, and the definition of operational failure. The probabilistic engine evaluates the failure definitions and accounts for uncertainties of model parameters. Furthermore, a quick screening module and the incorporation of existing deterministic calculation engines are described. Functional specifications for each module are given.

A staged development plan and underlying strategy are presented that outlines the step-wise development of CORA's functionalities. The plan identifies various go/no-go decision milestones and is flexible to re-rank functional improvements and change the sequence of certain development steps over the course of time. A pilot version of CORA has been developed to demonstrate some fundamental functionalities.

Examples are given focussing on fault leakage from a geomechanical point view. The example is using the Geertsma calculation engine assuming pressure depletion in a hypothetical disc-shaped reservoir. The function of the quick screening module is demonstrated using a Geomec model that includes some 40 potential fault leakage pathways.

Four Appendices provide more detailed information about the geomechanical workflow in the ORP, the Assessment Tree and failure mechanisms, the Geertsma calculation engine implementation and the software design of the pilot implementation of CORA.

Acknowledgements

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1. Introduction

1.1. Business objective

Carbon Capture Storage (CCS) is mentioned as the second of six strategic pathways for Shell to help addressing the world's CO₂ challenge [1]. Capacity, Injectivity, and Containment are the three main technical topics that are addressed for safe, long-term and economical management for CO₂ storage in geological formations. Appropriate project decisions should mitigate identified risks and uncertainties to an acceptable level (ALARP). A storage project will not commence without satisfactory resolution of the risks associated with these topics, whereas Monitoring, Measurement & Verification (MMV) is required to monitor residual risk during the injection and post-injection period.

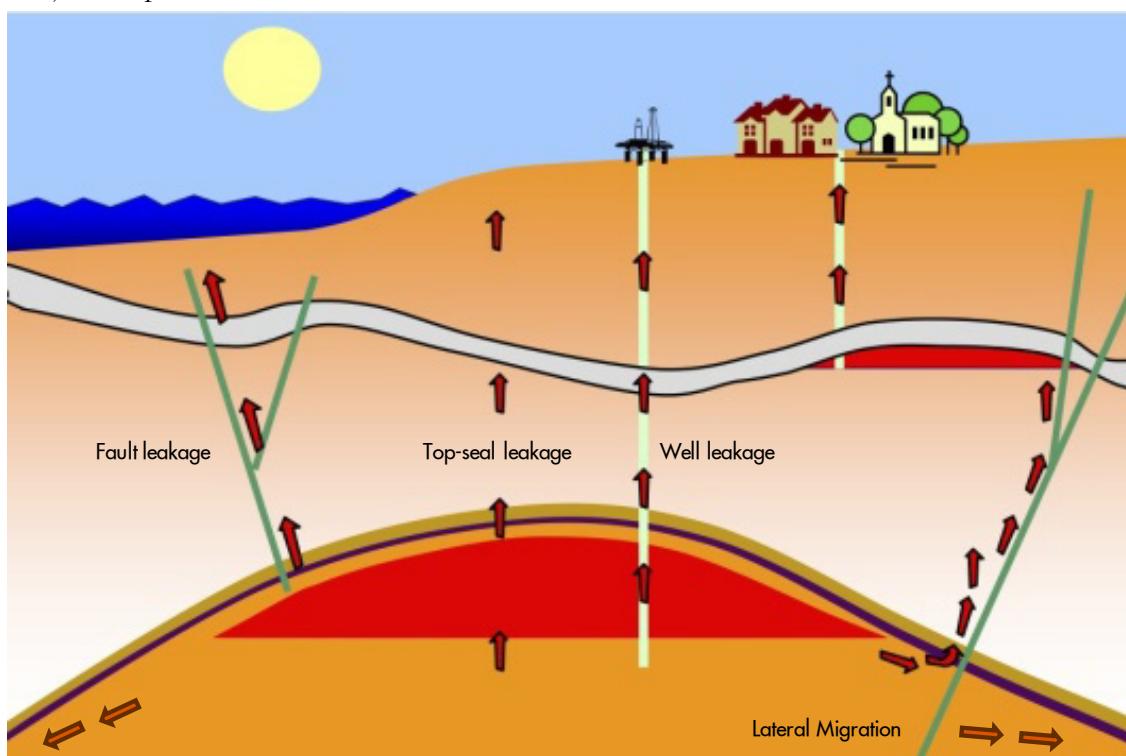


Figure 1.1 Fluid flow through top-seal, along wells, through fault zones and by lateral migration is the main leakage pathways from subsurface containers.

The risks and uncertainties associated with subsurface containment are indicated in Figure 1.1. Both, during the injection and post-injection stage of a project, containment of CO₂ is determined by the sealing function of the top-seal and the occurrence of potential leakage pathways along wells and faults. Furthermore, lateral migration may cause (reactive) fluids to leak out off the storage complex. The leakage pathways need to be considered individually and collectively when evaluating containment reliability. The objective of CORA is to provide a quantified assessment of the leakage risk from a storage complex.

1.2. Technical objective

Containment reliability is about estimating the likelihood that a subsurface container will operate according to agreed requirements within a specified time frame. To this end, in September 2008, a project was initiated with the stated goal to “Construct a PC-based tool to enable Shell’s global assets to assess the reliability of the containment of a field with respect to leakage to the surface or to connected sub-surface aquifers”.

Further specification included the following aspects:

- The reliability framework and GI-D PC tool for containment evaluation will be used as input into project wide risk models and to steer future R&D needs;
- The reliability framework and PC-Tool is to be used by experts and non-experts alike, and
- Should include (but not restricted to) a failure mode type analysis, the likelihood of geomechanical failure, hydrological leakages paths and leakage paths resulting from chemical degradation of wells;
- Enable the assessment of (additional) monitoring and surveillance data on the containment reliability of projects;
- The parameter input and associated uncertainties (e.g. probability distribution functions) should be supported by databases, analogues, analytical and numerical models. The input and updating of databases and uncertainty distributions should be accessible to Shell experts.

The objective of this report is to describe the design of the Containment Reliability Assessment tool CORA, to identify the key development issues, and demonstrate the concept using a pilot version with limited functionality focussed at geomechanical aspects. This work has been conducted in collaboration with Deltares, building on their experience with risk-based assessment of the Dutch water defence system. The work commenced early 2010.

1.3. Technical scope

Various trapping mechanisms [2], [3] need to be addressed when assessing subsurface containment of reactive fluids over an elongated period. Structural & Stratigraphic trapping is most important on an operational time scale, whereas mineral trapping of CO₂ is important on a geological time scale. Relevant other trapping mechanisms are residual trapping and solubility trapping. The interaction of CO₂ with subsurface solids and fluids needs to be understood to assess the relevance of various trapping mechanisms for a particular storage complex.

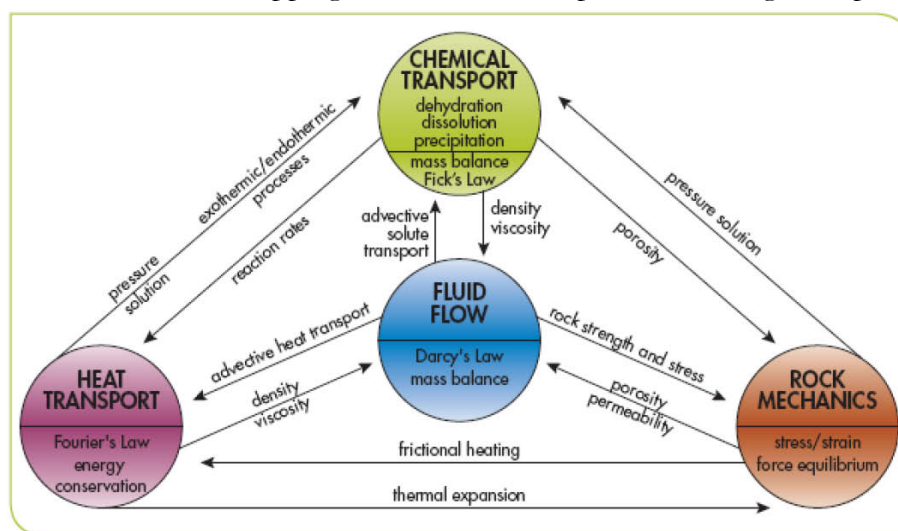


Figure 1.2 Overview of main processes in reactive flow modelling (modified from [4])

Reactive flow is the combined simulation of fluid, heat and chemical transport and the geomechanical response of the subsurface (see Figure 1.2, and references [3], [4]). CO₂ injection causes plume migration (transient flow) in the storage complex, triggers various chemical reactions with rock and fluids [5], [6], and induces geomechanical responses (see e.g. [7]). Reactive flow modelling describes the interaction of these processes and enables the assessment of various CO₂ trapping mechanisms, as well as capacity, injectivity and lateral migration issues of a storage complex. The relevance of various reactive flow processes is not fully understood yet,

and is subject of research globally. In particular the geomechanical aspects and their interaction with fluid, heat and chemical transport is relatively immature. It is the aim for CORA to address all reactive flow processes, either individually or in a coupled fashion. However, initial developments are focussed on the geomechanical aspects of containment.

1.4. Functional Requirements

CORA is designed to provide a probabilistic framework that follows “CO₂QUALSTORE, Guideline for Selection and Qualification of Sites and Projects for Geological Storage of CO₂” [9] to provide a quantification of containment risks. CORA results should integrate with TESLA, a tool used by teams to assist deciding risk mitigation options in development projects. Thereto, CORA should support TESLA’s assessment tree for CO₂ project and be able to evaluate all leakage pathways mentioned above.

Explicit, appropriate and commonly accepted operational failure definitions are required for leakage pathways individually and for the entire storage complex. Preferably, these should be based upon regulatory requirements. A listing of potential failure mechanisms is required that could cause leakage to occur. This implies a discrimination between material failure and operational failure (leakage). A quick screening of the most critical failure mechanisms and associated leakage pathways helps focussing the risk assessment and reduction of computational efforts.

CORA should be able to invoke various deterministic calculation engines to assess the identified failure mechanisms. These engine include geomechanical engines to evaluate the stress distribution in the subsurface, and a reservoir simulator to evaluate the reactive flow process. Various levels of modelling complexity should be possible.

The geomechanical evaluation approach needs improvement on two main aspects, namely a systematic approach to deal with uncertainties and a clear definition of failure (Figure 1.3). Geomechanical evaluations are currently conducted using discrete values for all model parameters, as indicated by “INPUT” in Figure 1.3. In this approach, best-guess values are taken to calibrate the model to available field data. Regularly, extreme values are selected for a number of model parameters to generate worst-case scenarios. The reservoir scenario under consideration is perceived safe, if the failure criterion is not met within a safe margin. However, the definition of failure or the safety margin is often not made explicit and reliability is therefore not consistent between projects.

Alternative probability distribution functions are required to describe the uncertainty of various model parameters. A normal distribution function is often a good initial assumption, but also uniform and log-normal distributions with truncation options are required to describe uncertainties realistically. Furthermore, a capability is needed to constrain uncertainties based on available field measurements. Geometrical uncertainties require a scenario approach in which alternative spatial (static) models are evaluated.

Efficient calculation algorithms are required to maintain evaluation time within practical limits. Monte Carlo simulations is a conventional approach to assess uncertainties, which involves the random sampling of model parameter from its probability distribution to produce numerous scenarios (also called iterations or trials, right hand column of Figure 1.3). This is not a feasible approach, because a single geomechanical model evaluation may take between a few hours and more than a day.

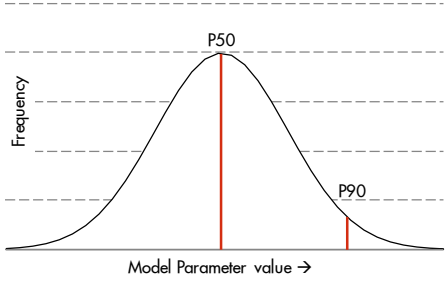
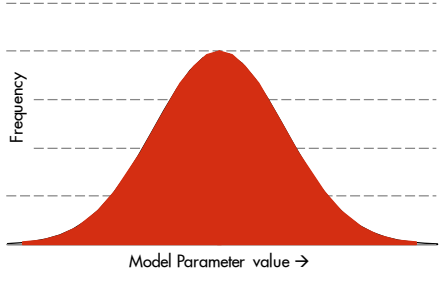
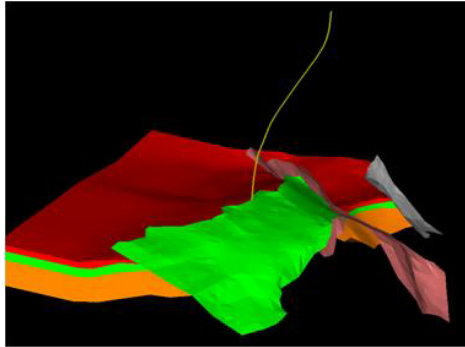
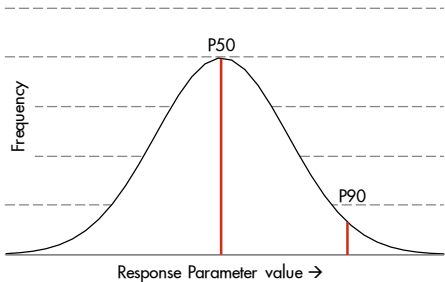
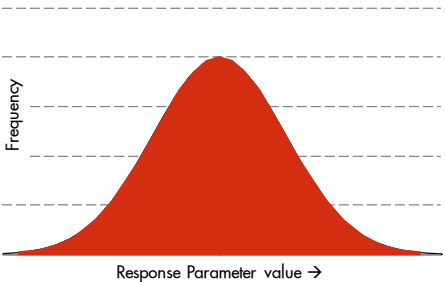
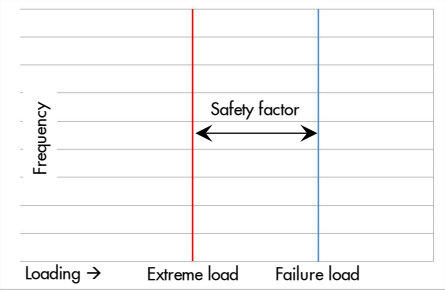
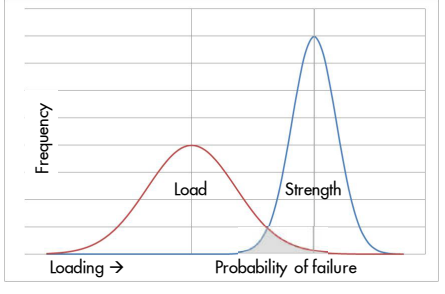
	Deterministic evaluation	Probabilistic evaluation
INPUT	 A limited number of scenarios are generated using best-guess (P50) or extreme values (P90) values for all model parameters.	 A large number of scenarios are generated by random sampling of model parameters from their probability distribution functions.
MODEL	 A model provides response parameters, such as maximum subsidence, fault permeability, maximum strain or fluid pressure along a well path	
OUTPUT	 Best-guess (P50) or extreme values (P90) are calculated for all response parameters.	 A probability distribution function is calculated for all response parameters
EVALUATION	 Margin or safety factor between strength and extreme loading condition	 Probability of failure is the likelihood that the loading exceeds the strength

Figure 1.3 Deterministic and probabilistic evaluation approach of geomechanical failure modes in the subsurface.

In summary, the following functionalities need to be in place to assess CO₂ containment reliability

1. Systematic approach to account for uncertainties
 - a. Approach to determine the distribution functions for all model parameters
 - b. Effective approach to deal with geometrical uncertainties
 - c. Efficient calculation algorithms to achieve practical calculation time for probabilistic evaluation
 - d. Algorithms to incorporate various sources of field data
2. Clear definition of failure
 - a. Definitions of operational failure for CO₂ containment in general and for individual leakage pathways in particular.
 - b. Assessment tree that identifies distinct failure mechanisms and their relation to individual leakage pathways
 - c. Quick screening approach to identify relevant failure mechanisms
 - d. Relationship between material and operational failure

1.5. Report structure

This report provides the design of the Containment Reliability Assessment framework CORA and a demonstration of some of the functionalities using a pilot implementation built in Microsoft Excel and Visual Basic for Applications. The pilot implementation was developed to evaluate the feasibility of most of the above listed functionalities. The strategy was to make available a very rudimentary version of CORA to demonstrate the concept and test feasibility quickly, and to improve elements subsequently, prioritised by the business.

In chapter 2, the context is provided in which CORA is aimed to operate by outlining the risk assessment issues and key deliverables for each phase in the Opportunity Realisation Process (ORP). The geomechanical evaluation workflow and modelling strategy is addressed in more detail in Appendix 1.

The modular design of the CORA framework is outlined in chapter 3. The assessment tree is given and the relation with TESLA is provided. Furthermore, the approach to material and operation failure is given. A more detailed discussion of various failure mechanisms is provided in Appendix 2.

The concept is demonstrated in chapter 4 by evaluating top-seal and fault shear failure reliability using the Geertsma nucleus of strain model. Details of the Geertsma engine and the response parameters used in the shear failure assessment are given in Appendix 3. A quick screening approach is provided using the same response parameters derived from results of the Geomec calculation engine. The details of the CORA pilot implementation are given in Appendix 4.

The development strategy and plan for the CORA framework is given in chapter 5, whereas conclusions and recommendations are given in chapter 6.

2. CORA in the ORP workflow

2.1. Workflow outline / CORA deliverables

CORA is aimed to fit in the ORP workflow of development projects [10] in general and the “CO₂QUALSTORE, Guideline for Selection and Qualification of Sites and Projects for Geological Storage of CO₂” [9] in particular. Therefore, key activities and deliverables for each ORP phase are taken as input to the functional specifications of CORA. The geomechanical activities for each ORP phase are outlined in more detail in Appendix 1.

The generic roadmap for CO₂ storage projects (Figure 2.1) has an upfront Screen Phase in addition to conventional hydrocarbon development projects. In this phase, multiple geographical locations with potentially multiple storage candidates are being considered. This is an important difference with the Identify Phase, in which - in principle - only one site candidate is being considered.

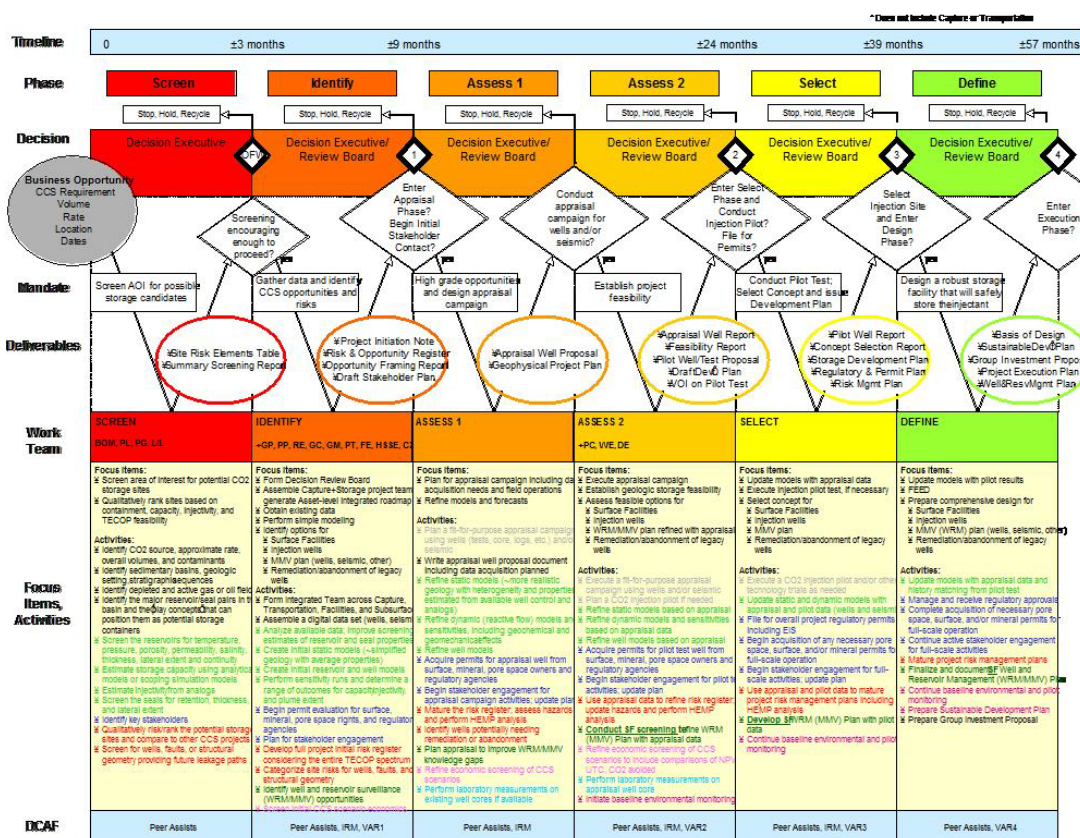


Figure 2.1 Generic CO₂ storage project roadmap [10].

2.2. ORP Screen risk assessment and deliverables

In the ORP Screen Phase, alternative storage options are ranked based on an estimate of the containment risk level. It is proposed to estimate the risk level using a “risk point model” incorporating key site characteristics such as the storage depth and the nature of the in-situ stress regime (compressive, extensional or strike-slip). These characteristics highly determine the pre-stress on faults and top-seal, and thereby the on-set of mechanical failure. The ranking using a “risk point model” should be supported by analogues and previous experience.

The storage options can be classified in a category of sites that have a favourable containment risk (green), sites for which the containment risk is inconclusive yet (amber) and sites that have unfavourable containment risk (red). CORA is probably not useful in this ORP phase due to the

qualitative nature of the risk assessment and the limited availability of site-specific (reactive flow or geomechanical) modelling data.

2.3. Risk assessment and deliverables in ORP Identify

In this phase, an assessment tree is agreed that specifies how various leakage pathways may contribute to the undesirable top-event. To this end, the CORA assessment tree may be pruned as appropriate. Thereto, a quick screening of potential geomechanical failure modes and reactive flow aspects are conducted. A probabilistic evaluation may not be appropriate yet, but a range of outcomes should be given in the form of a Tornado plot. Potential leakage pathways are identified for the storage options under consideration.

2.4. Risk assessment and deliverables in ORP Assess

The operational failure definition for the top-event and all underlying leakage pathways are articulated in ORP Assess, and should reflect best-practices and regulatory requirements. A probabilistic model is developed that includes appropriate (uncoupled) models for fluid, heat and chemical transport and geomechanical aspects, and uncertainty ranges of associated model parameters. Containment reliability (probability of failure) is evaluated against the definition of operational failure.

Deliverables included an overview of the most likely leakage pathway under alternative injection scenarios. If warranted, a “Value of Information” study should advise on data gathering options for an appraisal campaign, based on the expected uncertainty reduction of potential leakage pathways.

2.5. Risk assessment and deliverables in ORP Select

In ORP Select, better reactive flow simulations are pursued to improve the containment reliability assessment. This may comprise coupling of fluid, heat and chemical transport and/or geomechanical phenomena, increased subsurface detail, and the inclusion of non-linear (geomechanical) behaviour of formations. The focus is on generating and evaluating risk mitigation options in order to minimise the development of leakage pathways.

Deliverables are integral part of the Field Development Plan (FDP) and should be reflected in the dynamic reservoir pressure scenario, the design of injection (and production) wells and their safe operating envelope, and the design of monitoring equipment. Data acquisition frequency, follow-up interpretation and history matching activities are specified in the Well & Reservoir Management plan. Also, skills requirements and costs estimates are provided to execute the WRM plan.

2.6. Risk assessment and deliverables in ORP Operate

Deliverables are part of a collaborative effort of an Asset Team during the injection or post-injection stage. Deliverables focus on providing updated assessment of containment reliability and their influencing factors. In particular, the workflow should deliver proposals for additional data acquisition to reduce uncertainty and to provide input to asset team decisions, such as allowable injection (and production) pressure and rates, temperature and (CO₂) fluid composition.

3. CORA framework Design

3.1. Framework

The generic evaluation process for containment reliability comprise of five steps as indicated in Figure 3.1. Calculation Engines are deterministic software tools to analyse fluid, heat and chemical transport and geomechanical response of the subsurface due to CO₂ injection and storage, such as MoReS and Geomec. In principle any engine can be used, existing or new developments, as long as results can deliver the relevant response parameters by Post-Processing. Response parameters express the condition of a particular fault, well or top-seal as a function of time for the scenario under investigation. An optional Quick Screening of response parameters highlights the most critical leakage pathways in the subsurface and the sensitivity to uncertain model parameters. The function of a Quick Screening is to reduce the number of potential leakage pathways and uncertainty parameters for full probabilistic evaluation. The Probabilistic Engine calculates the probability distribution function of the response parameters based on the uncertainty range of the input parameters of the associated calculation engine. It evaluates the probability of failure of each well, fault or top-seal (component) and of the containment system collectively, using the specified operational failure definition.

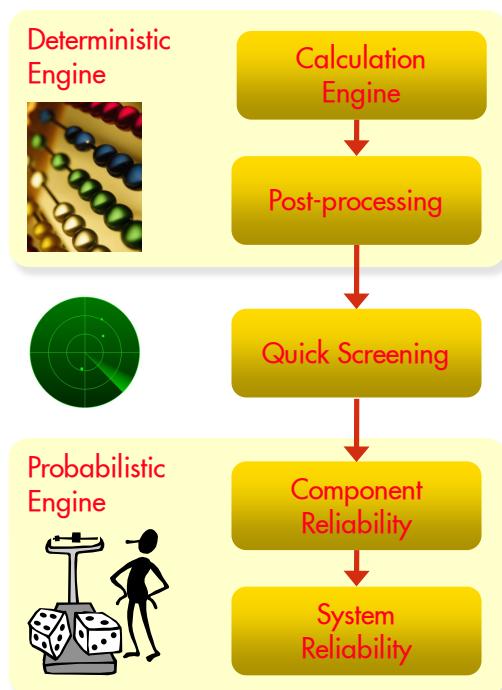


Figure 3.1 The CORA modular design for Containment Reliability Assessment makes use of exiting calculation engines to provide response parameters of faults, wells and top-seals for a components reliability assessment. An Assessment Tree facilitates the reliability assessment of the entire storage complex against regulatory requirement.

The probabilistic engine includes two essential features: an assessment tree and a set of operational failure criteria. The Assessment Tree specifies potential leakage pathways and failure modes in the subsurface that could lead to the undesired top-event. The Assessment Tree maps on the tree used in TESLA (see next section), and can be pruned or adjusted to reflect the project's bow-tie analysis. Operational failure criteria can apply to any event in the assessment tree. They are cast in a so-called limit-state function of response parameters, and should be made fit-for purpose for the project to reflect regulatory requirements.

The framework of Containment Reliability Assessment comprise of five modules, each taking care of one of the evaluation steps. The modular approach allows selection of appropriate tools for each ORP phase and incorporation of future improvements, such as coupled reactive flow modelling and regulatory requirements. The modules of the probabilistic engine are discussed in the next sections, whereas the deterministic modules, including the quick screening are discussed in chapter 4 using a geomechanical example.

3.2. Assessment tree

3.2.1. The TESLA Assessment Tree

An Assessment Tree, or more commonly, a Fault Tree Analysis (FTA) [11] analyses high-level system failures and all lower-level (sub-system) failure modes. System failure is represented by an undesirable top event at the root of the tree. All situations that could lead to the top event are added to the tree as a series of logic expressions. FTA is similar to the left side of a so-called Bow-tie analysis [19]. Alternatively, FTA is also referred to as Failure Tree or Event Tree Analysis. The neutral term Assessment Tree is used in this report.

FTA is considered a "top down" analysis, whereas a Failure Mode Effects & Criticality Analysis (FMECA, also FMEA) is considered a "bottoms up" analysis. FMECA and FTA are compatible methods of risk analysis, with the choice of method dependent on the nature of the risk to be evaluated. FTA is very good at showing how resistant a system is to single or multiple initiating failures. It is not good at finding all possible initiating faults. FMECA is good at exhaustively cataloguing initiating faults, and identifying their local effects. It is not good at examining multiple failures or their effects at a system level. FTA considers external events, whereas FMECA does not. For these reasons, FTA is the most appropriate approach for evaluating technical risks of a CO₂ storage complex.

The TESLA assessment tree in Figure 3.2 provides development teams with a systematic approach to mitigate technical risk of CO₂ storage projects. The technical risk aspects are part of the project wide approach to risk management that follows the TECOP principle (section 3.2.3 of [8]). The four elements of the tree coincide with the main project drivers - Capacity, Containment, Injectivity, and MMC (Monitoring, Measurement and Verification) - and ensures a comprehensive coverage of relevant factors and uncertainties.

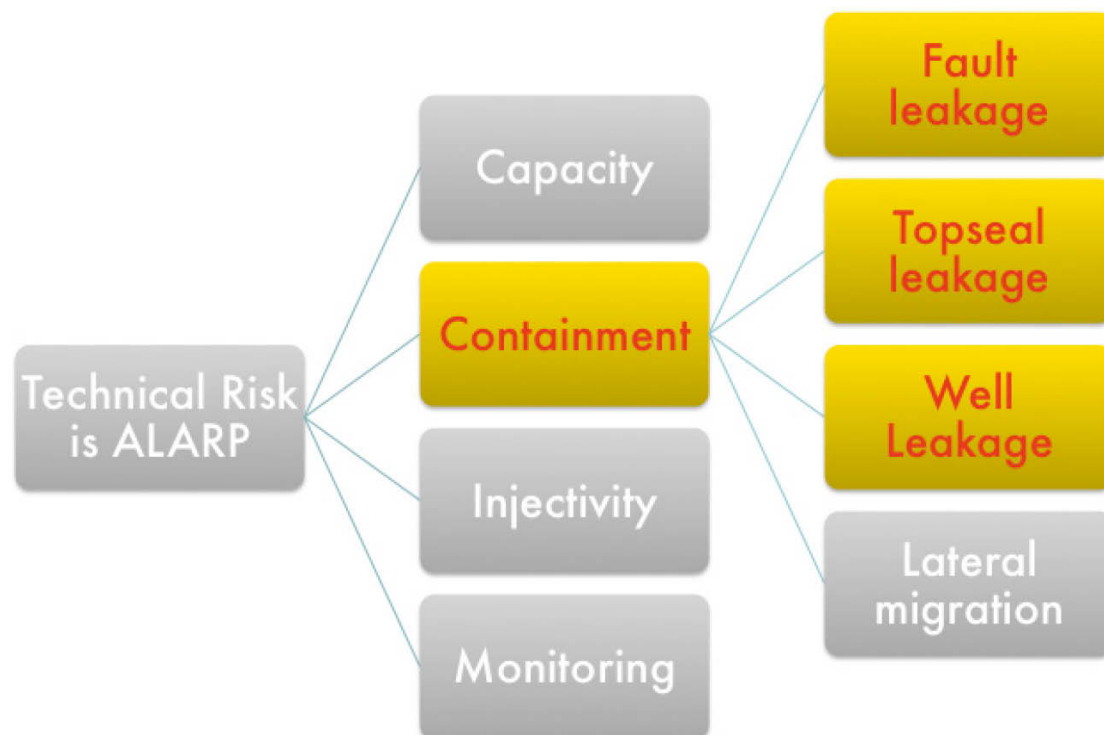


Figure 3.2 The first two levels of the TESLA tree. The scope of CORA is indicated by the yellow boxes.

TESLA uses Evidence Support Logic to verify or falsify the hypotheses in the Assessment Tree [12]. A so-called “Italian Flag” representation is used to express the latest judgment on every hypothesis. The available evidence either supports the hypothesis, contributing to the “green side” of the Italian flag, or rejects the hypothesis and thereby contributes to the “red side” of the flag. Absent or inconclusive evidence is represented by the “white space” in the middle. The objective is to reduce uncertainty for every hypothesis by the collection of evidence, with a neutral attitude to its outcome.

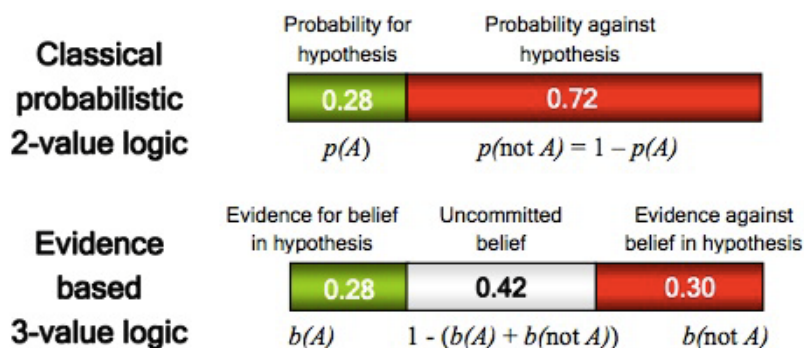


Figure 3.3 Classical two-value probabilistic analysis compared with three-value interval analysis, after Egan [12].

Evidence can be provided through field and lab measurements and through (modeling) studies. TESLA assists in the selection of project work to be conducted in order to reduce the level of uncertainty, and bring down project risk to a level As Low As Reasonably Practicable (see section 4.6.6 of the OPMG for demonstrating ALARP). TESLA generates an Evidence-ratio plot (Figure 3.4) to visualise the available evidence for each leaf hypothesis in the assessment tree and track progress made during project maturation.

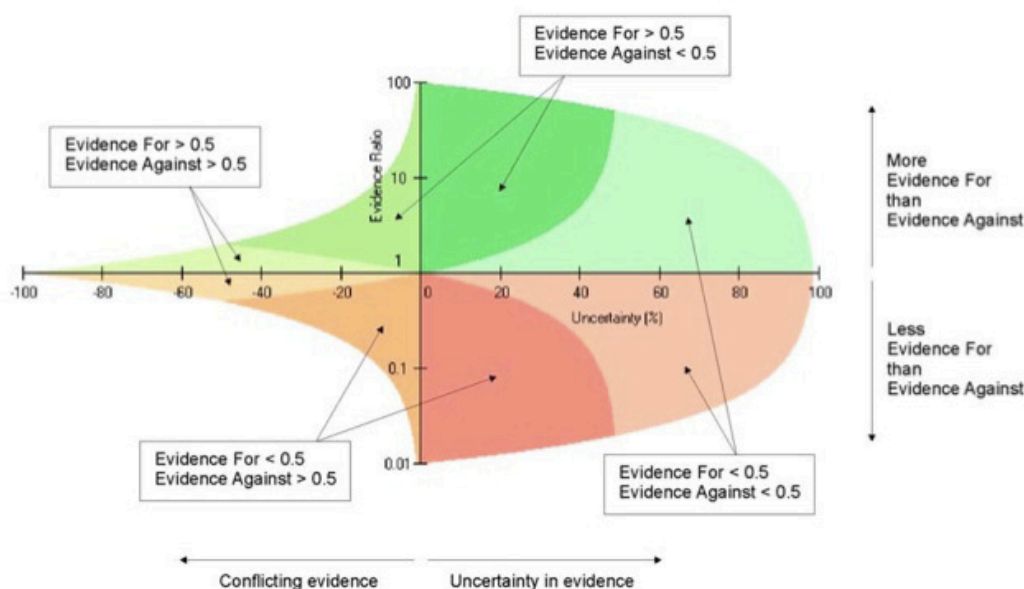


Figure 3.4: Regions in the Evidence-ratio plot [12].

3.2.2. The CORA Assessment Tree

The objective of the Containment Reliability Assessment framework CORA is to provide quantitative evidence to the TESLA hypotheses through probabilistic evaluation of the leakage risks. Thereto, the framework in Figure 3.1 includes probabilistic capabilities while using existing calculation.

A preliminary outline of the leakage risks to be evaluated (Appendix 2) is largely based on the Barendrecht Bow-Tie analysis [13], [14]. An extensive overview of long-term containment risks is available, e.g. through Quintessa's online Features, Events and Processes (FEP) database [15]. This database focuses on the long-term safety and performance of the storage system after the injection stage. Further work is required to identify the potential leakage risks to be evaluated in the injection stage of projects and to specify the evaluation workflow in terms of data acquisition and geomechanical and reactive flow modelling.

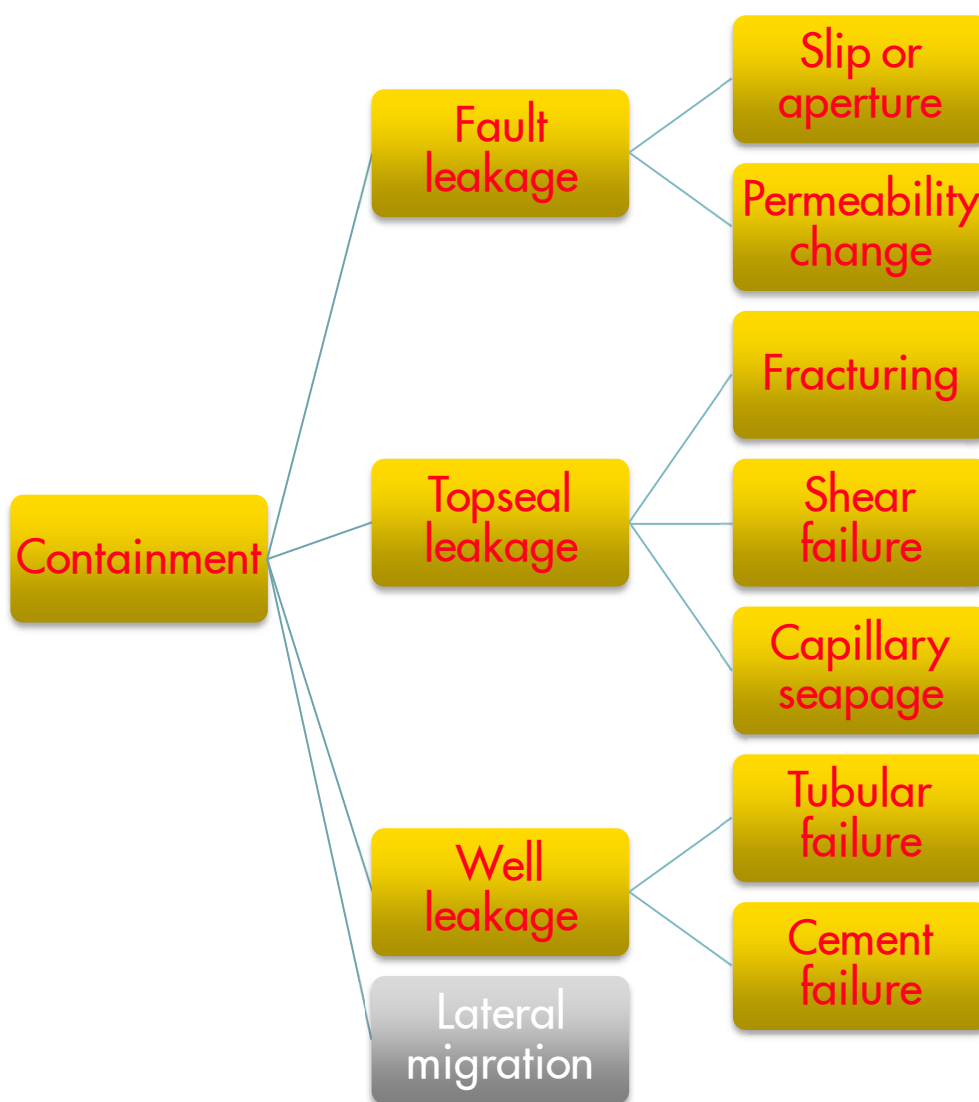


Figure 3.5 The CORA Assessment tree maps on the TESLA tree. Lateral migration is not covered by CORA in this stage, but can be included by updating the assessment tree and inclusion of associated calculation engines.

3.3. Operational Failure

3.3.1. *Top event and global failure*

The top event in the CORA assessment tree defines operational failure of a CO₂ container using a quantitative expression that discriminates between acceptable leakage and not-acceptable leakage (failure). Alternatively, also a 3-value logic as shown in Figure 3.3 could be adopted, requiring two quantitative expressions to discriminate between the categories failure, uncertain and no failure. Preferably, these formulations are founded on regulatory requirements. CORA is flexible to evaluate any formulation of operational failure as long as the underlying calculation engines provide the algorithms to deliver the response parameters.

According to the IPCC definitions [2], CO₂ leakage is considered not significant - if “the fraction retained in appropriately selected and managed reservoir is very likely to exceed 99% over 100 years, and is likely to exceed 99% in 1000 years”. Here, “very likely” is interpreted as a probability between 90% and 99%. This definition is focused at the key function of storage facilities to securely remove CO₂ from the atmosphere on a global scale.

3.3.2. *Failure at branch level - local failure*

CORA also allows definition of operational failure at lower level branches of the assessment tree. That is, operational failure may also be declared if the release of CO₂ from the containment complex is exceeding a predefined limit for a specific fault, well or top-seal, or a group thereof. This may be relevant if leakage into a particular aquifer or other subsurface formation is to be restricted.

IPCC has also articulated some constraints regarding hazards affecting people or the environment in the direct vicinity of the leakage. A concentration greater than 7-10% in air would cause immediate dangers to human life and health. Furthermore, leakage of CO₂ or displacement of brines into aquifers is mentioned as risks. Exceeding these limits would constitute local failure. If available, regulatory requirements of the local operating environment apply.

3.3.3. *Limit-State Function*

The operational failure definitions, either using a 2-value or 3-value logic, are represented by limit-state functions. The functions relate the expected value for CO₂ release from the containment complex to the agreed limit value (limit-state value). A proper estimation of the CO₂ release through the top-seal, along well trajectories and along fault planes requires complex reactive flow calculations. Quite a few of the reactive flow processes are currently insufficiently understood. For the time being, it may therefore be more practical to express operational failure in other measures than in CO₂ release volume or rate. In CORA the limit-state function can therefore also be expressed in thermal, chemical or geomechanical parameters to define operational failure. More generally, these parameters are referred to as response parameters.

The slip section of a fault is an example of a geomechanical response parameter to specify operational failure. The area of the fault for which the actual shear stress is exceeding the carrying capacity defines the fault slip section, and can be expressed for instance in m² or fraction of the fault area. In the first place, such a failure definition implies that fault slip can be related to CO₂ leakage. It is assumed that permeability along a fault plane increases if (a part of) the fault slips. This will not be true in all cases. Secondly, it assumes that some fault slippage is allowed before operational failure occurs (before the limit-state function is satisfied). All assumptions are practical substitutes to arrive at an assessment of operational failure and can be used in CORA.

3.3.4. Probability of failure

CORA evaluates the probability of operational failure taking into account the uncertainty of model input parameters of the calculation engines. This can be done for the limit-state function associated with the top event on system level, or those on sub-system level. In fact, the limit-state function evaluates the actual loading on a system in relations to its operational limit (strength) as indicated in Figure 3.6.

A safety factor can be defined in both deterministic and probabilistic evaluations. In probabilistic evaluations the safety factor is defined as the margin or ratio between the expectation (P50) value for the loading and the strength. The safety factor does not change if the uncertainty of either factors change, but the probability of failure does. In a deterministic evaluation, the safety factor is the only way to express the margin between load and strength, because the distribution functions of both are not taken into account.

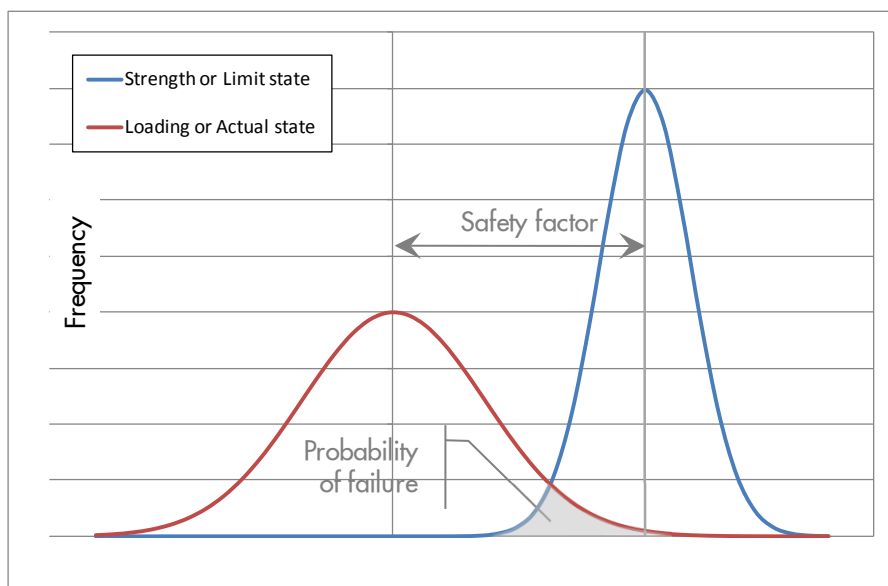


Figure 3.6 Failure is at stake if the loading exceeds the strength of a system. The Limit-State Function is an evaluation of this relationship. The safety factor expresses the margin between loading and strength in a deterministic approach, whereas the probability of failure can be calculated if their distribution functions are known.

4. Geomechanical Reliability Assessment

4.1. Calculation Engines

Calculation Engines are deterministic software tools to analyse fluid, heat & chemical transport and geomechanical response of the subsurface due to CO₂ injection and storage. In principle any engine can be used as long as results can deliver the relevant response parameters by dedicated post-processing. Geomechanical calculation engines that evaluate the subsurface response to a reservoir pressure and temperature change include Geertsma, QuickBlocks and Geomec. Other engines are available to evaluate specific well related problems such as FIST (sand production) and CASINT (casing integrity). Here, Geertsma is briefly described to demonstrate the type of results expected from a calculation engine. Additional detail is provided in Appendix 3.

A Geertsma model of the subsurface [18], [20] assumes a horizontal disc-shaped reservoir with radius R and height h at a depth D . The overburden and reservoir are assumed to have the same linear-elastic properties and a homogenous in-situ stress condition. Any difference in horizontal stress is not accounted for due to the axi-symmetrical nature of the model. Finally, a homogenous reservoir depletion ΔP (negative for inflation) is assumed. Table 4.1 provides an overview of all model input and output parameters.

Table 4.1 Input and output parameters for the Geertsma model and values of an example Base Case.

Reservoir geometry	Symbol or Label	Base Case	Unit
Reservoir depth (top)	D	2000	[m]
Reservoir height	H	100	[m]
Reservoir radius	R	2000	[m]
Reservoir pressure			
Depletion pressure	ΔP	5	[MPa]
In-situ Stress			
Vertical stress gradient	$S_{v\text{-grad}}$	22,5	[kPa/m]
Horizontal stress gradient	$S_{h\text{-grad}}$	15,8	[kPa/m]
Pore pressure gradient	$P_{p\text{-grad}}$	10,5	[kPa/m]
Formation Properties			
Compressibility	C_m	$1,7 \cdot 10^{-4}$	[1/MPa]
Poisson's ratio	ν	0,25	[-]
Evaluation Point			
Coordinates Point P	$\mathbf{x}_p = (r_p, z_p)$		[m]
Output Parameters			
Displacement vector	$\mathbf{u}_p(u_r, u_z)$		[m]
Stress vector	$\sigma_p(\sigma_{rr}, \sigma_{\theta\theta}, \sigma_{zz}, \sigma_{rz})$		[MPa]

The displacements and stress changes are calculated in a single point P outside the disc-shaped reservoir according to the Geertsma nucleus of strain model. The elastic strain tensor ϵ^{el} can be calculated by taking the first order derivative of the displacement tensor with respect to the coordinates r , θ and z . The stress and displacement routines are called repetitively for the evaluation of e.g. the strain distribution along a well trajectory or the stress distribution on a fault plane. This analytical approach is implemented in Microsoft Excel's Visual Basic (Appendix 3).

Alternatively, the QuickBlocks or Geomec calculation engines can be used to generate the output parameters in Table 4.1. These engines allow taking more realistic assumptions for the static subsurface model, while Geomec can also calculate the plastic strain ϵ^{pl} according to a wide range of plasticity models, the thermally induced strain ϵ^{th} , and many more geomechanical aspects of the subsurface. These engines calculate the requested data for the entire subsurface model at once rather than in a single point as in the Geertsma model.

4.2. Post-Processing

The objective of post-processing is to calculate response parameters that are appropriate representations of the loading condition of faults, top-seal and well sections. Leakage pathways may develop if the carrying capacity is exceeded and mechanical damage is incurred. Shear and tensile failure are key mechanical failure mechanisms highlighted in the assessment tree. Here, attention is focussed on four response parameters related to shear failure, which can be used in the assessment of leakage pathways along fault planes and through the top seal. Response parameters for other failure modes can be articulated in a similar way.

The Shear Capacity Utilisation (SCU) in a material point is defined as the actual effective shear stress τ divided by the maximum allowable shear stress τ_{max} :

$$SCU = \frac{\tau}{\tau_{max}} \quad (4.1)$$

A material point is at the onset of shear failure if $SCU=1$, whereas a value smaller than 1 indicates intact material with a linear-elastic response. Values larger than 1 can be calculated if linear-elastic behaviour is assumed, indicating that the material point is beyond its carrying capacity and a plasticity model, such as the Mohr-Coulomb model discussed in Appendix 3, should be employed.

4.2.1. Geomechanical response parameters for top-seal evaluation

Four response parameters (Table 4.2) are introduced to provide representative information about the shear loading condition in the top seal when using linear-elasticity theory. All four parameters are evaluated along a pre-defined two-dimensional surface in the top seal, to be chosen just above the reservoir. The surface can have an arbitrary shape, but is assumed flat and horizontal in the example below. The average SCU provides an indication of the remaining shear capacity left in (a part of) the top seal. Shear stress will be transferred to neighbouring rock if the shear capacity is exceeded in a particular point. This is accounted for in a proper non-linear geomechanical assessment. The average SCU may act as an approximation of the shear capacity left after redistribution.

Table 4.2 Geomechanical response parameters for shear failure assessment along a pre-defined surface in the top seal.

Response Parameter	Description
Average SCU	Average shear loading condition in the top-seal and
Maximum SCU	Shear load level in the most critical point in the top seal
Failure area [m ²]	Area of the pre-defined surface in [m ²] for which $SCU \geq 1$
Failure area [%]	Area of the pre-defined surface in [%] for which $SCU \geq 1$

The maximum SCU provides information of the most critically loaded point in the top-seal. The failure area will only be larger than zero if the maximum $SCU \geq 1$, thus if any shear failure is calculated. Generally, some material failure can be allowed before the integrity of the top-seal is at stake. A limit-state function based on the maximum SCU from a linear-elastic evaluation can be a first approximation of one based on a non-linear evaluation. This type of approach is also followed to determine the mud weight for bore hole stabilisation [16]. The failure area in [m²] may be a proper response parameter to estimate the leakage rate through the top seal based on a relationship between shear failure and permeability.

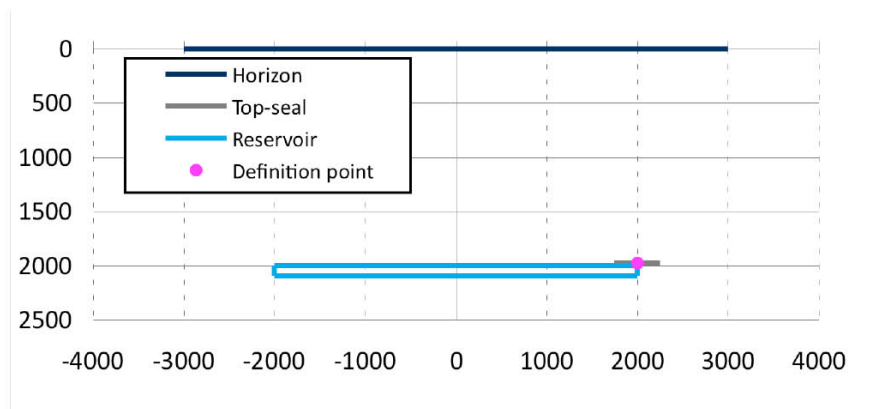


Figure 4.1 Top-seal evaluation around the corner of a disc-shaped reservoir (Geertsma model) at 1980m depth.

The four response parameters listed in Table 4.2 are evaluated along a 500m portion of the top-seal at 1980m depth (Figure 4.1) for the Base Case values in Table 4.1. To this end, the following calculations are conducted at some 100 points along the indicated top-seal section:

- the stress vector using the Geertsma calculation engine (Appendix 3)
- the principal stress components
- $\tau / \tau_{\max}$ (see Appendix 3)
- the response parameters

The result is shown graphically in Figure 4.2 for 5 MPa reservoir depletion pressure. The Mohr-Coulomb cohesion C is 5 MPa and friction angle ϕ is 22 degrees. The maximum SCU value of 0.89 is found just outside the depleted reservoir. Table 4.3 summarises the response parameters for 4 different depletions stages. The onset of shear failure at this surface is found at a depletion pressure of about 6.5 MPa.

Table 4.3 The response parameter values for shear failure in the top-seal at 4 depletion stages.

Depletion	Shear Capacity Utilisation (SCU)		Failure length	
	Average	Maximum	[m]	pct
5 MPa	0,631	0,893	0	0%
6 MPa	0,639	0,963	0	0%
7 MPa	0,647	1,035	15	3%
8 MPa	0,657	1,108	30	6%

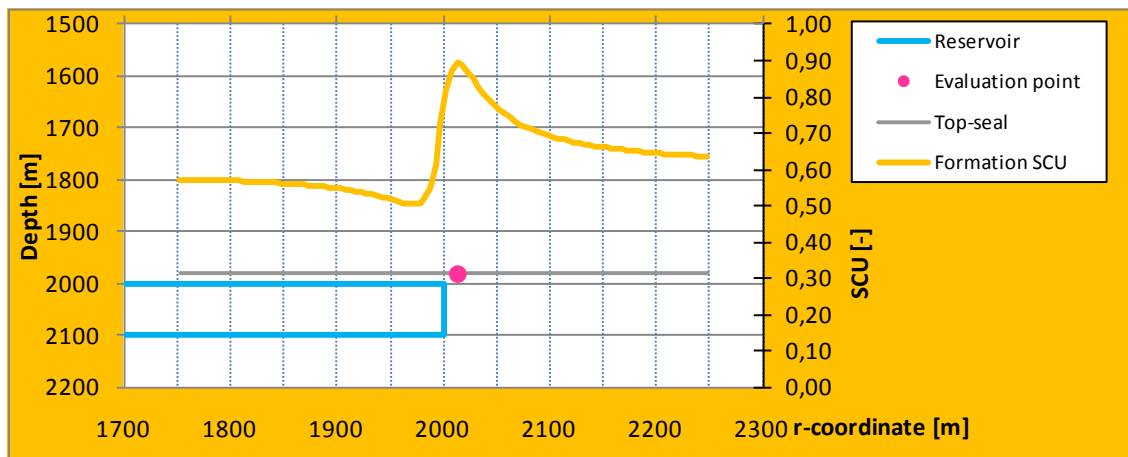


Figure 4.2 The Shear Capacity Utilisation according to Geertsma's nucleus of strain model along a surface in the top-seal at 1980m depth (grey line) and at 5 MPa depletion (top) and the associated (bottom) for a cohesion of 5 MPa and a friction angle of 22 deg. The maximum SCU value is 0.89 (right-hand scale).

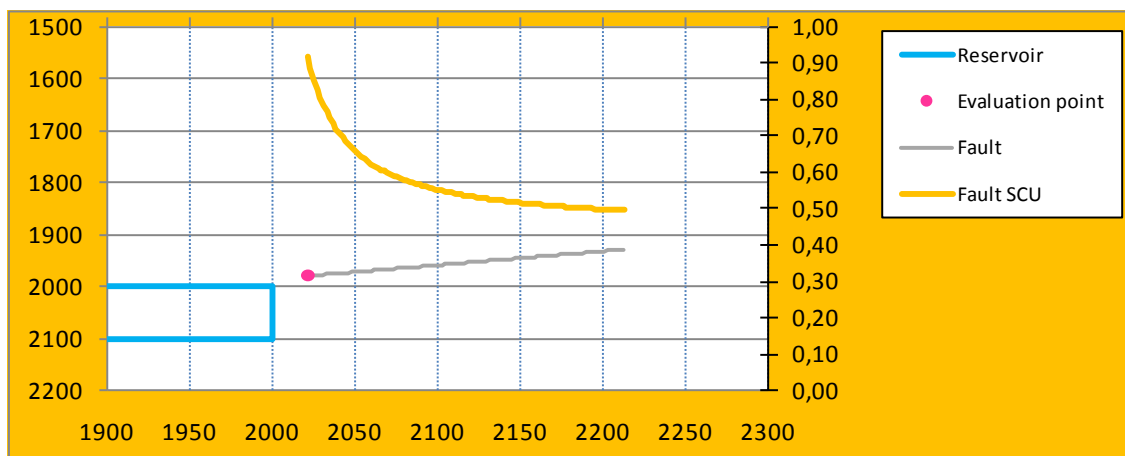
4.2.2. Geomechanical response parameters for fault evaluation

The response parameters for fault slip evaluation are similar to those for shear failure in the top-seal and given in Table 4.2. The distribution of the Shear Capacity Utilisation along a fault plane is plotted in Figure 4.3. The fault is making an angle of 105 degree with the downward vertical axis (anti-clockwise) and starts above the edge of the depletion volume at 1980m depth. The evaluation length is 200m and the Mohr-Coulomb cohesion C and friction angle ϕ are taken somewhat lower than for the top-seal at 2 MPa and 13 degrees respectively. The Geertsma model of the subsurface is taken as in Figure A3.1.

The SCU is calculated using the normal and shear stress components at the angle of the fault plane. This is somewhat different from the calculation of the SCU for the top seal, which is based on the principal stress components. From Table 4.4 it is seen that no fault slip occurs at 5 MPa depletion pressure, because the maximum SCU remains smaller than 1. The average SCU remains well below 1 at increasing reservoir deletion, even though a slip section is developing gradually. This implies that the remaining shear capacity of the entire fault is still significant. A non-linear analysis would be required to estimate the fault slip section more accurately.

Table 4.4 The response parameter values for shear failure a fault plane at 4 depletion stages.

Depletion	Shear Capacity Utilisation (SCU)		Failure length	
	Average	Maximum	[m]	[pct]
5 MPa	0,573	0,916	0	0%
6 MPa	0,597	1,005	2	1%
7 MPa	0,620	1,093	6	3%
8 MPa	0,643	1,180	10	5%

**Figure 4.3** The Shear Capacity Utilisation along a fault plane (grey line) at 5 MPa depletion according to Geertsma's nucleus of strain model. Cohesion is 2 MPa and friction angle is 13 deg. The maximum SCU value is 0.92 (right-hand scale).

4.3. Quick screening

The objective of a quick screening is to find the most critical location in the subsurface with respect to the development of leakage pathways, and to provide focus to the probabilistic evaluations. The quick screening presented in this section is not essential for a probabilistic assessment. Attention is restricted to a geomechanical quick screening for fault slip, using response parameters proposed in the previous section.

Table 4.5 shows the output of a quick screening of a Geomec model with some 40 fault planes. The data was provided through a SVS¹ file that contained the triangulated fault surfaces. The number of triangular segments in each fault plane is given in column 3, and the total area is given in the last column of the table. Furthermore, the SVS file contained the normal and maximum shear stress for 3 depletion stages calculated by Geomec. The quick screening was conducted for a cohesion of 4 MPa and a friction angle of 20 degrees for all fault planes and a single depletion stage. Finally, the table was sorted on the fraction of the fault that is slipping, and next for the average SCU of the fault plane.

It is seen that 10 faults are slipping over the entire area for the given cohesion and friction angle. The next 10 faults are partly slipping, whereas the remaining 20 stay in the elastic regime. In the last category, 15 fault planes have an average SCU < 0.35. It seems reasonable to assume that these faults can be excluded from probabilistic evaluation for the assumed depletion scenario.

¹ Shell proprietary geological software tool to model fractures, faults and sweet spots

Alternative depletion scenarios could be evaluated in a similar fashion to find the fault planes that are most critically stressed in all cases, and focus the probabilistic evaluation accordingly.

Table 4.5 Quick screening of faults in a Geomec model based on the Shear Capacity Utilisation (SCU, eq. (1)). The maximum and average SCU are given for each fault plane in conjunction with the fault area slipping (SCU>1) in m² and as fraction of the entire fault area.

Name	Type	Seg mts	SCU avg.	SCU max	Failure area [m2]	Failure fraction	Total area
StrucMod_July_Rly_4bc_trim_ts	fault	48	3,90	3,99	87858	1	87858
StrucMod_July_Flt_42_trim_ts	fault	80	3,88	4,03	62033	1	62033
StrucMod_July_Flt_03b_trim_ts	fault	35	3,79	4,01	398867	1	398867
StrucMod_July_Flt_10a_trim_ts	fault	28	2,36	2,47	1140190	1	1140190
StrucMod_July_Flt_41_trim_ts	fault	33	2,11	2,32	352917	1	352917
StrucMod_July_Rly_16_trim_ts	fault	40	1,43	1,51	325104	1	325104
StrucMod_July_Flt_10b_trim_ts	fault	36	1,20	1,23	452141	1	452141
StrucMod_July_Flt_30_trim_ts	fault	19	1,12	1,20	2016050	1	2016050
StrucMod_July_Flt41_b_trim_ts	fault	19	1,06	1,09	341389	1	341389
StrucMod_July_Flt_08_trim_ts	fault	20	1,02	1,05	89282	1	89282
StrucMod_July_Flt_05_trim_ts	fault	107	3,20	3,68	712302	0,999	713113
StrucMod_July_Flt_03_ok_trim_ts	fault	69	2,51	2,64	631382	0,989	638313
StrucMod_July_Flt_10_trim_ts	fault	33	1,06	1,16	1006634	0,715	1407845
StrucMod_July_Flt_14_29A3_trim_ts	fault	50	0,93	3,30	422761	0,402	1051734
StrucMod_July_Flt_50_trim_ts	fault	158	0,83	1,27	796703	0,315	2532054
StrucMod_July_Rly_50_trim_ts	fault	165	0,66	2,52	208917	0,226	925282
StrucMod_July_Flt_22_trim_ts	fault	34	0,41	3,03	124558	0,197	631894
StrucMod_July_Flt_19_trim_ts	fault	140	0,48	3,51	48178	0,181	266873
StrucMod_July_Flt_04_trim_ts	fault	122	0,33	1,48	172918	0,104	1670144
StrucMod_July_Flt_north_0_trim_ts	fault	82	0,92	1,20	174052	0,058	3003606
StrucMod_July_Flt_04b_trim_ts	fault	40	0,68	0,70	0	0	179230
StrucMod_July_Flt_30a_trim_ts	fault	25	0,63	0,68	0	0	905357
StrucMod_July_Flt_16_trim_ts	fault	59	0,62	0,98	0	0	1544803
StrucMod_July_Flt_56_trim_ts	fault	33	0,53	0,57	0	0	341820
StrucMod_July_Flt_23_trim_ts	fault	18	0,52	0,53	0	0	184139
StrucMod_July_Flt_48_trim_ts	fault	35	0,33	0,38	0	0	262765
StrucMod_July_Flt_55_trim_ts	fault	28	0,24	0,26	0	0	290254
StrucMod_July_Rly_8a_trim_ts	fault	53	0,23	0,26	0	0	139613
StrucMod_July_Flt_54_trim_ts	fault	93	0,22	0,26	0	0	676065
StrucMod_July_Flt_17_trim_ts	fault	23	0,20	0,40	0	0	645294
StrucMod_July_Flt_70_trim_ts	fault	38	0,14	0,16	0	0	518191
StrucMod_July_Flt_25_trim_ts	fault	42	0,12	0,15	0	0	234796
StrucMod_July_Flt_40_trim_ts	fault	18	0,09	0,11	0	0	604729
StrucMod_July_Falt_15_trim_ts	fault	124	0,09	0,16	0	0	612816
StrucMod_July_Rly_16a_trim_ts	fault	88	0,07	0,08	0	0	784343
StrucMod_July_Flt_4C_trim_ts	fault	25	0,05	0,07	0	0	78616
StrucMod_July_Flt_12_trim_ts	fault	30	0,03	0,11	0	0	2433258
StrucMod_July_Rly_9a_trim_ts	fault	130	0,03	0,41	0	0	318882
StrucMod_July_Flt_09_trim_ts	fault	151	0,03	0,07	0	0	1091621
StrucMod_July_Flt_11_trim_ts	fault	26	0,02	0,04	0	0	2310393

4.4. Reliability assessment

A fast probabilistic geomechanics workflow developed by Pao [22] has been implemented in Microsoft Excel's Visual Basic for Applications (VBA, see Appendix 4) to assess containment reliability of individual leakage pathways. In essence, the method develops a so-called proxy-function for a particular response parameter that substitutes a computational intensive engine in the probabilistic assessment. A proxy-function is an algebraic expression that relates the uncertain model parameters to one response parameter. Currently, a linear series expansion is used for the proxy functions.

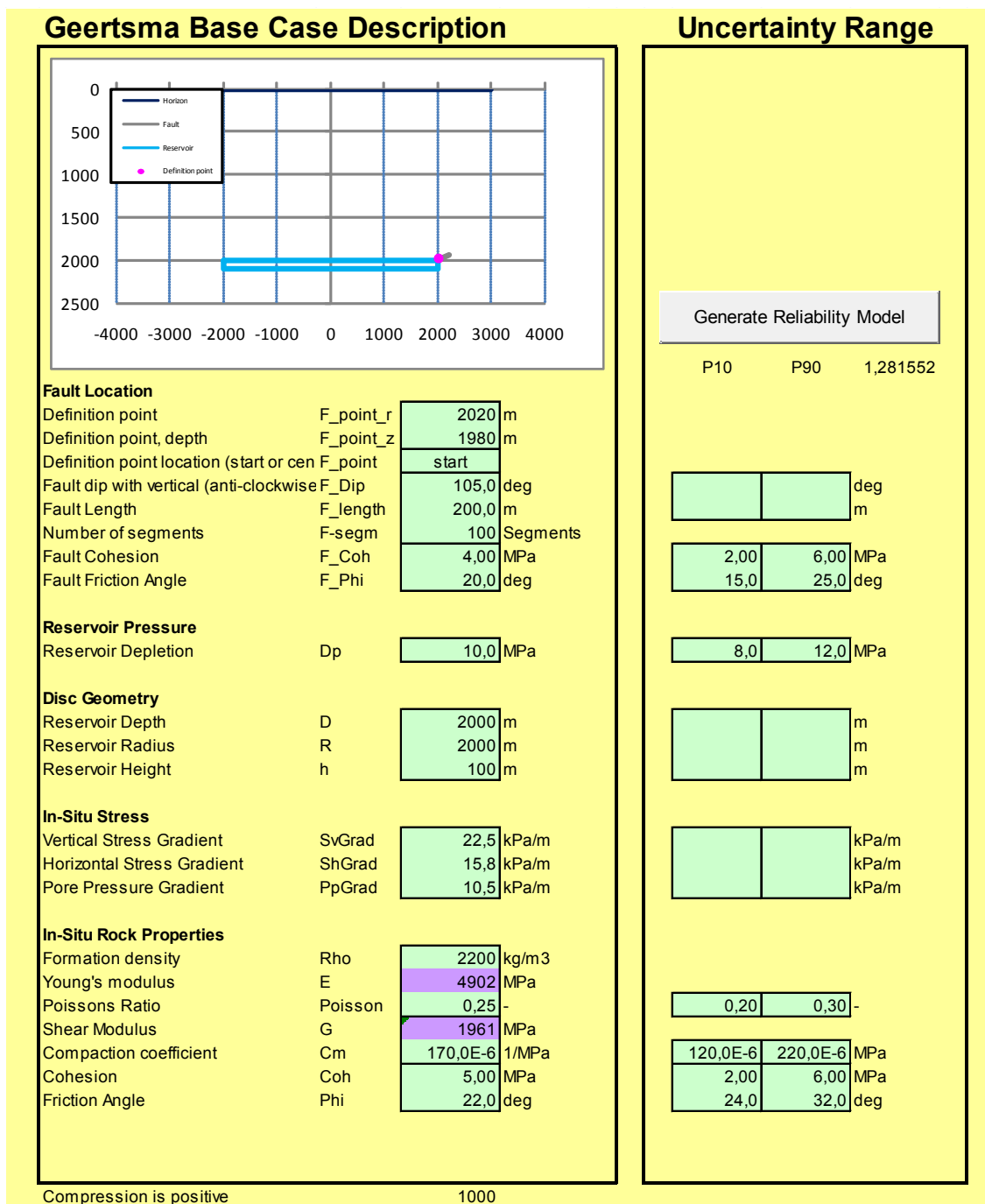


Figure 4.4: The Geertsma model and uncertainty ranges for selected model parameters to evaluate fault reliability

To demonstrate the method, the fault reliability in a Geertsma model of the subsurface is considered. The same model is used as in section 4.2.2 with additional uncertainty ranges for 7 model parameters as given by their P10 and P90 values in Figure 4.4. The impact of each uncertain model parameter on the response parameter is displayed in a Tornado chart (Figure 4.5). The extreme values for each uncertain model parameter in the Tornado chart are obtained by substitution of its P10 and P90 values in the proxy function, while keeping all other parameters at their P50 value. The Tornado chart in Figure 4.5 is symmetrical around the P50 or expectation value of the maximum SCU, which is about 0.83. Extreme values reach about 1, the onset of shear failure at the fault plane. The coefficients in the proxy function represent the sensitivity of the response value to each model parameter (see Appendix 4).

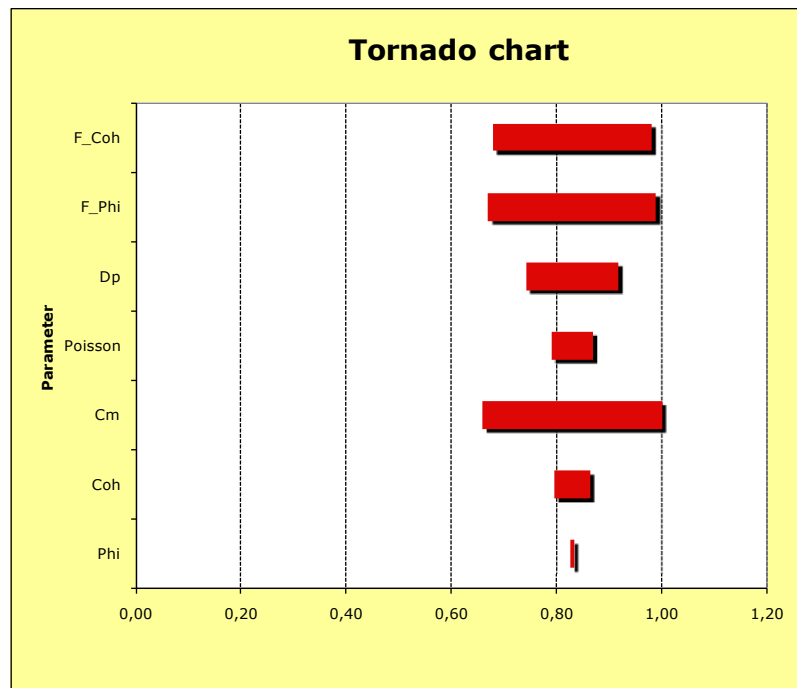


Figure 4.5 Tornado diagram showing the sensitivities of model parameters on the Fault's maximum SCU. The uncertainties are based on the uncertainty ranges provided (see Figure 4.4).

The reliability and probability of failure using the fault's maximum SCU is calculated in Figure 4.6. The expectation value for this response parameter is 0.83 (Figure 4.5) and is based on the average values of the uncertain model parameters given in Figure 4.4. The probability that the maximum SCU will exceed a (limit-state) value of 1.5, 1.75 and 2.0 for a given reservoir depletion pressure is given in the right-hand graph of Figure 4.6. The graph of the left-hand side gives the same result in terms of the reliability index. It is obvious that the most conservative failure definition yields the highest probability of failure and the lowest reliability. In Figure 4.7, the reliability and probability are calculated if failure is defined by the average SCU exceeding 0.6, 0.8 and 1.0 respectively. It is noted that the influence of the reservoir depletion pressure is difficult to discern for low probabilities.

The reliability index provides a better way to discern the difference between alternative values for the reservoir pressures and the limit-state values. The reliability index is defined as R/r [22], where R is the distance from the mean value of the response parameter to the limit state value, and r is the distance from the mean value to the so-called design point. The design point is the combination of model parameters with the largest probability of failure and is not known a priori. The reliability index is 1 if the design point coincides with the limit state value (onset of failure).

A 3-value logic can be employed to support the “Italian Flag” concept used in TESLA, by the evaluation of alternative limit-state values of the response parameter as given in the figures. A “green” could be assigned if the reliability index is larger than, say, 2 for a conservative criterion, whereas a “red” is given if the reliability index is smaller than 2 for an aggressive criterion. The key is to find acceptable criteria and threshold values that are commonly accepted by project stakeholders.

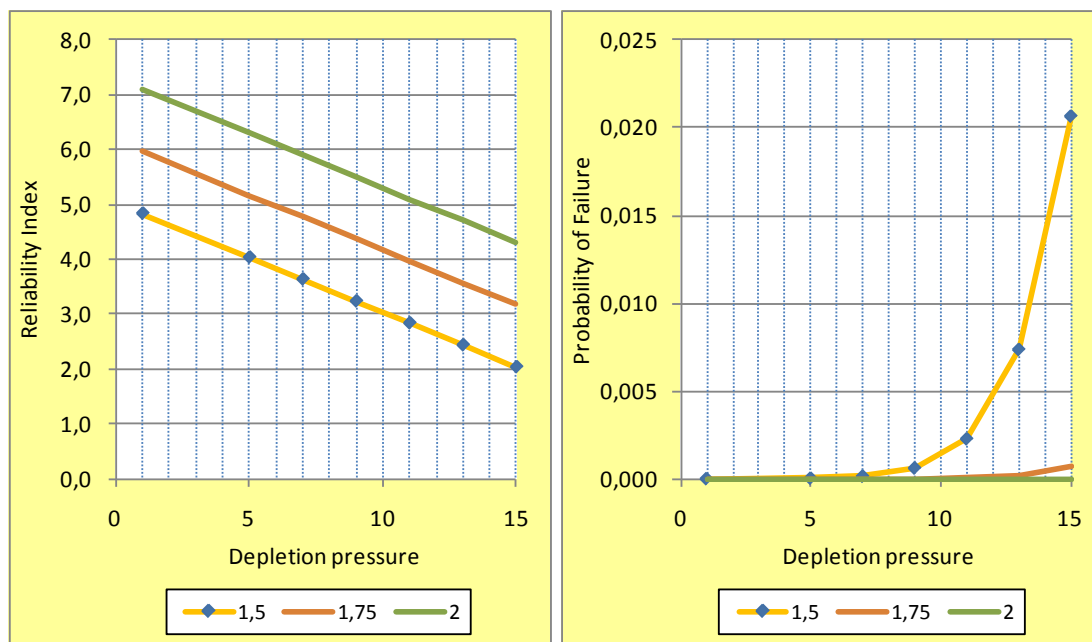


Figure 4.6 The reliability index (left) and the probability of failure (right) based on 3 different limit values for the maximum Shear Capacity Utilisation on the fault plane. The deterministic results are in Table 4.4.

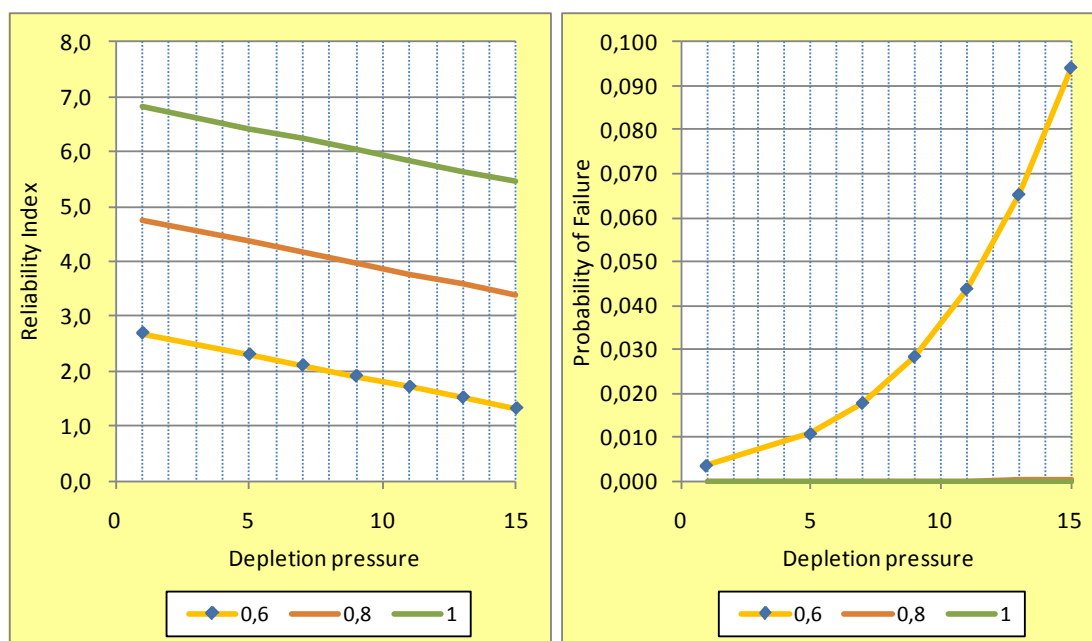


Figure 4.7 The reliability index (left) and the probability of failure (right) based on 3 different limit values for the average Shear Capacity Utilisation on the fault plane. The deterministic results are in Table 4.4.

5. CORA development plan

5.1. Staged Development of Cora

The development of a containment reliability assessment tool includes technical and non-technical challenges. One of the over-arching uncertainties is whether or not the desired containment reliability can be achieved given the large uncertainty of the subsurface response to CO₂ injection. This includes uncertainty of relevant reactive flow processes, which is subject of R&D, as well as uncertainty of model parameters. Another main uncertainty is the usefulness and acceptance of Quantitative Risk Analysis (QRA) internally within Shell's development projects and externally.

A development approach is followed to reduce uncertainty associated with the acceptance of QRA in the design process of CO₂ storage projects. Therefore, the initial approach is to develop a rudimentary tool that can demonstrate how QRA can contribute to project decision-making, rather than building a tool that has many of the identified functionalities. Although most of the underlying technologies are known, the proposed workflow and tool can be regarded as “emerging” solution for which “proof of concept” still needs to be achieved (see Figure 5.1).

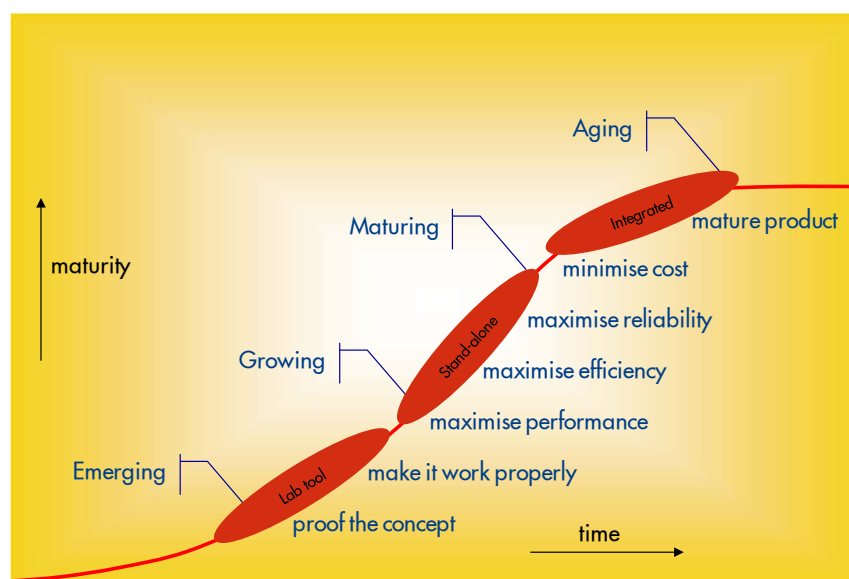


Figure 5.1 Phase 2 of the CORA project aims at developing a lab tool to conduct a proof of concept

The software that embodies the CORA framework comprise of three parts:

1. The CORA calculation kernel, which includes all on-board calculations such as the probabilistic evaluations (see Figure 3.1),
2. The external deterministic calculation engines, such as MoReS and Geomec, and
3. The graphical user interface (GUI), which facilitates the transfer of data between various input sources, external calculation engine and the CORA calculation kernel, and allows fit-for-purpose interpretation of the results by the user.

Different occurrences of the user interface may exist as the supported workflow (data, analysis, results) of users may differ. However, the CORA calculation kernel and external calculation engines should be the same and should therefore be portable between hardware platforms as appropriate. The three development stages in Figure 5.1 are described in the next sections.

5.2. Lab tool

The lab tool is for specialists to provide proof of concept and to test the approach on realistic case studies (see Figure 5.1). This version of the tool focuses on providing probabilistic functionalities. Therefore, key objective of this stage is to develop a re-usable CORA calculation kernel and test relevant probabilistic approaches. At least two major releases of the lab tool are foreseen: one with basic functionalities and a second release with more advanced functionalities. This second release may include approaches to constrain the probabilistic evaluation with field observations and to extend the deployment scope of the tool by improving computational efficiency. Experimental Design – a technique to reduce computational efforts - has already been evaluated [22] and will therefore be included in the early release. Current expectation is that the CORA calculation kernel needs re-development if migrated into the integrated tool.

The user interface of the lab tool will assume “Excel-level” skills of the users. Linear-Elastic (LE) geomechanical calculation engines (Geertsma, QuickBlocks, Geomec-LE) are the first to be incorporated. Accuracy of results is not considered as key deliverable in the initial phase of the design. Input and Output (IO) functionalities of incorporated calculation engine are used as much as possible. IO related to the probabilistic calculations are provided through the CORA user interface. The user interface will be abandoned in the stand-alone stage.

5.3. Stand-alone tool

The development of the stand-alone tool will focus on the graphical user interface for deployment by (trained) project engineers in an independent fashion. Requirements for input and output, analysis options, speed, and interfaces with deterministic calculation engines should be obtained from the lab tool stage. Interfaces with (operations) databases and other analysis result will be matured in this stage. A web-based user interface with server-client architecture is foreseen as it minimises dependency of the user interface and computational kernel.

The stand-alone tool will have a full suite of deterministic calculation engines to address reactive flow processes relevant for CO₂ storage in the subsurface. Only mature calculation engines will be incorporated that have a proven track record.

Calculation efficiency of the probabilistic analysis will be matured. Techniques other than Experimental Design may be piloted in the lab tool and deployed in the stand-alone version. Calculation efficiency is regarded as an enabler for deployment of probabilistic analysis, because computational efforts are expected to become unpractical for relatively small subsurface models already. Selection and implementation of the most promising acceleration techniques is expected in this development phase of CORA.

5.4. Integrated tool

The development of an integrated tool will focus on incorporating coupled deterministic calculation engines. This could comprise of some form of coupled geomechanical & reservoir simulation (reactive flow simulation). Operational failure criteria would probably need reassessment, whereas a uniform probabilistic approach for reservoir and geomechanical assessment needs to be implemented. Re-development of the reliability assessment tool as part of a single system (DRMS/SRMS) would likely be required.

Table 5.1 Proposed development of various functionalities over the three tool generations.
Abbreviations: linear-elastic (LE), non-linear (NL).

	Lab tool pilot	Lab tool	Stand-alone user tool	Integrated tool (SRMS/DRMS)
Year	2010	2011	Next stage	Final
Workflow	None	Dev. & accept for geomech.	Dev. & accept for reactive flow, deploy geomech.	Deploy reactive flow, improve geomech.
System Reliability	None. No operational failure crit.	Geomechanical aspects only	Reactive flow & Geomechanics uncoupled	Reactive flow & Geomechanics coupled
Assessment Tree	Paper, proposal	Draft, implemented	Tested & improved	Mature version
Component Reliability	Shear failure of Top-seals & Faults only	All failure mech. of wells, top-seals & faults	Reactive flow & NL failure mech. of wells, top-seals & faults	Reactive flow & NL failure mech. of wells, top-seals & faults
Quick Screening	For material failure location	For material failure location	Leakage pathways & failure location	Leakage pathways, failure location, NL behaviour & coupled processes
Post-processing	Shear failure	Geomechanics aspects (LE) of wells, top-seals & faults	Reactive flow & geomech. aspects (NL) of wells, top-seals & faults	Reactive flow & geomech. aspects (NL) of wells, top-seals & faults
Calculation Engines	Geertsma	Geomec (LE) QuickBlocks Geertsma	Reactive flow & uncoupled geomechanics (LE&NL)	Reactive flow & coupled geomechanics (LE&NL)

Table 5.2 Deterministic geomechanical calculation engines for shear failure evaluation

	Geertsma	QuickBlocks	Geomec	
Geometry	Single disc	Multiple blocks	Realistic	Realistic
In-situ stress	Homogenous, constant gradients	Layered with constant gradients	Realistic	Realistic
Formation properties	Homogenous, linear-elastic	Layered, linear- elastic	Arbitrary, linear- elastic	Arbitrary, non- linear
Reservoir scenario	Single delta P	Delta per block	Realistic	Realistic

6. Findings & Recommendations

The fast probabilistic geomechanics workflow [22] has been successfully deployed for shear failure in top-seal and along fault planes in a pilot version of CORA. To this end, four response parameters have been tested and found useful and appropriate to monitor the shear failure condition when using a linear-elastic calculation engine. The approach is fast: the Geertsma nucleus of strain model evaluates a single scenario in about a second on a standard GI-D laptop computer, and a fault reliability assessment is done within half a minute. Geometrical uncertainties can be assessed with the Geertsma model with the same effort as other model parameters. However, the applicability for realistic field conditions may be limited due to axis-symmetrical nature of the model. Finally, the same response parameters have been proven useful for a quick screening of critically loaded faults using a linear-elastic Geomec model.

Acceptance of a quantitative risk assessment (QRA) methodology in CO₂ storage projects is critical to the deployment and further development of CORA. The technical probability of success of CORA has increased significantly after development of the pilot version of CORA. Exposure to active development projects is required to further de-risk this technology solution. Key issue is how to translate probability of failure into (better) project decisions. The results and interpretation in a probabilistic assessment are different than in a conventional deterministic approach and demand a different way of working. Additional data is required (distribution functions) and the definition of failure needs to be clearly defined. This is not always the case in geomechanical analysis.

CORA development strategy recommendations:

- Demonstrate CORA capabilities to support project decision-making using simple geomechanical calculation engines.
- Evaluate current limited CORA capabilities on active development projects and agree deployment plan, articulate functional requirements and prioritise developments.
- Demonstrate CORA using (reactive) flow calculation engine in early stage, and test potential linkage of geomechanical and reactive flow response parameters in limit-state function.

A strategy for further development of the CORA lab tool has been outlined in chapter 5. Specifically, it is recommended to

- Develop and test system level reliability based on (geomechanical) response parameters for well, top-seal and fault failure mechanisms.
- Review and accept the Assessment Tree as a basis for containment risk assessment in projects. This involves modelling specifications and associated field data acquisition.
- Develop, evaluate and accept definitions for operational failure (with external stakeholders ultimately) for each leakage pathway individually and for the containment system.
- Define and test (geomechanical and reactive flow) response parameters for all failure mechanisms in the assessment tree to allow evaluation of all leakage pathways.
- Extend and test the Quick Screening for additional response parameters and leakage pathways.
- Assess the Geertsma nucleus of strain model for screening of geometrical uncertainties in the early ORP phases.
- Identify and adopt evaluation models and associated calculation engines for all leakage pathways and failure mechanisms for incorporation into the CORA framework.
- Develop, evaluate and accept an approach to determine the probability distribution functions of all (linear-elastic geomechanical) model parameters.

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Appendix 1. ORP Workflow for containment assessment

The high level workflow is described for the technical assessment of containment risks for various phases of the ORP. The workflow is focused on, but not restricted to, the geomechanical aspects of containment risks. The following workflow elements are described when relevant:

- Study Objective
- Risk analysis
- Field data gathering
- Geometrical modelling
- In-situ stress modelling
- Formation property modelling
- Dynamic reservoir scenario modelling
- Evaluation options
- Deliverables

A1.1. ORP Screen

In the ORP Screen Phase, containment risk of multiple sites or geological storage options are classified by making use of analogues and empirical insights to help ranking. Fluid, heat & chemical transport or geomechanical aspects are considered separately, but without developing site-specific models. Instead, databases of previous experience may be used to support evaluation of the characteristic parameters of the sites under consideration with those of analogues.

The key geomechanical characteristics that impact site ranking comprise of geometrical aspects, in-situ stress, formation properties and the dynamic reservoir pressure. The reservoir depth and the nature of the in-situ stress regime (compressive, extensional or strike-slip) highly determine the current state of stress of faults and top-seals. The volume and magnitude of the inflation or depletion area are the main parameters describing the additional loading induced by the planned CO₂ or H₂S storage volumes. Formation types (clays, sandstones, carbonates or salt domes) determine the short-term and long-term geomechanical response of the subsurface. Furthermore, the number of wells and faults is a key measure of the number of potential leakage pathways that may be created. Evaluating these geomechanical characteristics do not require developing a time-consuming site-specific model that calculates the stress distribution in the subsurface.

An estimate of the containment risk level should be used for a ranking of alternative geological storage options. The risk level could be estimated using a “risk point model” incorporating the aforementioned characteristics. The storage options may be classified in a category of sites that have a favourable containment risk (green), sites for which the containment risk is inconclusive yet (amber) and sites that have unfavourable containment risk (red), because their characteristics with known containment issues red (not ok, drop).

CORA should not be deployed to support site selection in this phase of the ORP. CORA is aimed at a quantitative assessment of risks, while a qualitative assessment is more fit-for-purpose for this phase. Therefore, a “risk point model” is proposed, in which points are accumulated for site characteristics that are not favourable from a containment point of view.

A1.2. ORP Identify

A1.2.1. Objective & Deliverables

Objective of the geomechanical assessments in this ORP Phase is to identify potential leakage pathways in the subsurface (see Figure 1.1) that may jeopardize containment of CO₂. Discrete

failure locations are reported for the given structural and stratigraphic scenarios. Tornado plots should be provided for each potential leakage pathway (failure location) capturing uncertainties for various modelling parameters (geometrical, in-situ stress, formation properties) and control options (RE scenarios or other constraints). The objectives for further (geomechanical) assessment of containment risks are articulated.

A1.2.2. Data gathering

Available data will be gathered and filtered for relevance in the Identify phase. Large amounts of data may be available for producing assets, whereas only analogues or near-field data may be available for green field reservoir candidates.

A1.2.3. Risk Analysis

The CORA risk assessment tree is pruned and made fit-for-purpose for the project. The assessment tree specifies the undesirable top-event and all relevant underlying geomechanical failure modes. Associated scenario evaluations, modelling needs and data requirements are captured in the modelling strategy for the entire ORP. Ideally, this should be based on a qualitative risk analysis (e.g. Bow-tie analysis). In ORP Identify, the focus should be given to geometrical uncertainties, because these become increasingly expensive to assess in later phases. Input to the project risk register and Risk-Assessment Matrix (RAM) should be provided.

A1.2.4. Geometrical model

A simplified 3D structural representation of the reservoir and overburden is taken to allow deployment of analytical solutions if possible, or alternatively, a linear-elastic QuickBlocks model. A 2D representation should be considered if the geological structure allows so and does not jeopardize the main conclusions. This may be the case for simple geological structures and in the absence of faulting.

A1.2.5. In-situ stress modelling

A homogenous in-situ stress and pore pressure condition should be assumed. Vertical in-situ stress can usually be estimated relatively accurately based on analogues or integration of near-field density data. Tectonic setting and calliper log data provide input to the ratio of the two horizontal stress components relative to the vertical stress. Orientation of horizontal in-situ stress should also be assessed. A rotated in-situ stress orientation should be included already in the Identify phase, in case of a nearby salt dome or in case of shallow injection reservoirs with significant surface topography.

A1.2.6. Formation property modelling

The analytical solutions assume linear-elastic formation properties, which could be based on analogues from other fields or correlations functions if log data is available that is sufficiently representative. Stiffness contrast between reservoir and overburden may be considered, but homogeneity should be the starting point.

A1.2.7. Reservoir scenario modelling

Uniform pressure depletion and inflation scenarios are taken throughout the injection reservoir. The depletion/inflation volumes in the geomechanical model should resemble those used by the reservoir engineer (material balance). Shape of the depletion/inflation volume may deviate in this stage. Coupling of the reactive flow processes is manually, if any. Alternative reservoir units may

be considered in the geomechanical modelling, but only one at a time. Superposition of results may be considered if desirable.

A1.2.8. Evaluation Options

A quick assessment of potential failure locations and leakage pathways and their key influencing factors should be conducted. A probabilistic evaluation is not required, but a range of outcomes should be given in the form of a Tornado plot. The evaluations should focus on structural uncertainties, e.g. induced by time-depth conversions, such as reservoir depth and fault inclination.

The conclusions of the previous Screening phase, if available, are related to the new findings. A re-evaluation of the alternative sites may be advised if prior conclusions are rejected and site ranking might be significantly altered.

A1.3. ORP Assess

A1.3.1. Objective & Deliverables

The Assess phase may be separated into two parts: 1) business case for and design of an appraisal campaign, 2) interpretation of appraisal data and evaluation of alternative development concepts. The objective of the first part is to assess the “Value of Information” and determine a data acquisition program, whereas the objective of part 2 is to provide a preliminary quantitative Containment reliability assessment, highlighting the most likely leakage pathways for the development scenario under consideration.

The key difference between the two parts is that existing (nearby) field data and core material is used prior to the appraisal campaign, whereas new, field specific data and fresh core material is used in the post-appraisal campaign evaluations. In principle, the same (geomechanical) model should be used throughout the Assess Phase. The first part of this Phase may merge with the Identify Phase discussed in the previous section, if the candidate reservoir is (part of) a producing asset for which the structural uncertainty is already sufficiently reduced.

A1.3.2. Data gathering

Field data gathering should be aimed at uncertainty reduction of the identified containment risks. The modelling and evaluation efforts in the first part of Assess – as described below – should deliver a “Value of Information” document that advises on the data acquisition needs. This report should also demonstrate the impact of new field data on the containment risks in relation to estimated data acquisition costs.

A1.3.3. Risk Management

The Bow-tie analysis and assessment tree are refined with relationships between geomechanical and operational failure conditions. For example, operational failure may be defined in terms of the maximum (plastic) strain in the top-seal, or the maximum fault slip or aperture. Also, fluid, heat & chemical transport conditions may be included, such as the increase of fault permeability with the increase of slip section or the change of top-seal permeability and capillary entry pressure under altered stress condition and chemical reactions. Regulatory requirements with respect to the definition of operational failure should be incorporated at this stage.

A1.3.4. Geometrical model

Volume and shape of the (disposal) reservoir should be linked to the dynamic reservoir model. Depths and dip angles uncertainties of reservoir and non-reservoir formations should have been

reduced significantly by improved seismic time-depth conversions. It is assumed that this is available at the start of the Assess Phase. Appropriate representation may require the use of either QuickBlocks or Geomec depending on the reservoir geometry. QuickBlocks should be considered for slightly dipping reservoirs or slightly curved anti-cline structures whereas Geomec should be used for more complex geometries.

Additional formation layers should be included in the model to represent stiffness contrasts in the over- and under-burden, as well as in the reservoir. Inclusion of additional layers is driven by the need to better simulate the stress redistribution in the subsurface. Faults are not included in the model, but their behaviour is assessed during post-processing. The geometrical model, comprising of the reservoir and formation layers, may need updating after the appraisal campaign.

A1.3.5. In-situ stress modelling

P10, P50 and P90 values for the in-situ stress conditions are developed using correlation functions and available log data. These values should be provided for each formation or reservoir layer in the model. The uncertainty range is narrowed and in-situ stress inhomogeneities - both in terms of value and orientation - are incorporated once appraisal data – such as LOT data in reservoir and overburden - is available.

A1.3.6. Formation property modelling

P10, P50 and P90 values for the linear-elastic formation properties are developed using correlation functions and available log data. These values should be provided for each formation or reservoir layer in the QuickBlocks or Geomec model. Attention should focus on improved modelling of the formation stiffness contrasts, which comprise of Young's modulus and Poisson's ratio. Linear thermal expansion (of the solid phase) should be specified if significant thermal strains are expected.

Attention may focus on stiffness ratios between the formation layers prior to the data acquisition campaign. The ratio of Young's moduli fully determines the stress redistribution between formation layers. The absolute value determines the magnitude of displacement and strain. The latter is important when dealing with subsidence, or well failures. Stiffness ratios are appropriate for stress-based failure criteria, such as fault slip or top-seal fracturing. The correlation functions are therefore very suitable prior to the acquisition of fresh core material, as they are relatively good in estimating the stiffness ratios between formation layers. Rock mechanical core tests should calibrate and improve the absolute value of linear-elastic (and thermal) model constants.

A1.3.7. Reservoir scenario modelling

A uniform reservoir injection or depletion pressure should be assumed within the reservoir volume (tank model). Alternative reservoir pressure scenarios should be investigated, corresponding to the cases evaluated by the reservoir engineer. Thermal aspects should be included as required. It is vital that the strain loading, either induced by reservoir pressure, temperature or chemical changes is represented correctly.

A1.3.8. Evaluation Options

The evaluation effort is focused on the identification of the most likely leakage pathways under alternative reservoir pressure scenarios, and at reducing the associated uncertainties. To this end, coupled or uncoupled simulations of fluid, heat & chemical transport and geomechanical failure are conducted, taking into account uncertainty ranges of associated model parameters. Most

critical leakage pathways and the influencing factors are determined based on an assessment of containment reliability (probability of failure).

The (mechanical) failure condition of individual wells, top-seals and faults are evaluated for the provided uncertainty ranges of aforementioned model parameters. To this end, the stress and strain data is extracted from the geomechanical model along virtual well trajectories and fault locations. The 1st order fault slip magnitude is assessed as a range of possible outcomes, if re-activation appears to be an issue.

In the first part of the Assess Phase, evaluation is focused at the ranking of uncertainties supported by Tornado charts, whereas the emphasis is shifted to a preliminary assessment of containment reliability and the impact of potential mitigation options once appraisal data has become available. Improved containment reliability provides a measure for the “Value of Information” obtained containment provides a measure of the value of information for individual field measurements.

A1.3.9. Deliverables

Part 1 of the ORP Assess Phase should deliver a “Value of Information” document that advises on the data acquisition needs, whereas part 2 should advise on operational reservoir pressure limits to ascertain containment for a range of reliabilities, and for alternative development concepts. Input is provided to updating project risk register and Risk-Assessment Matrix (RAM). Geomechanical modelling advice is given for the ORP Select Phase in term of inclusion of geometrical features (formation layers, faults), in-situ stress anisotropy & orientation, additional lab tests for non-linear properties (on available core) and associated modelling. No further geomechanical evaluation is advised if all identified risks appear to be negligible. Also, technology mitigation options should be advised for further reduction of containment risks, which may comprise of monitoring requirements during the project Operate phase.

A1.4. ORP Select

A1.4.1. Objectives & Deliverables

The objective of the ORP Select Phase is to help selecting a development concept that optimises project value, while bringing leakage risks to a level as low as reasonably practicable (ALARP). Technically, the objective is to help selecting well locations, trajectories and their safe operating envelope, optimisation of the dynamic reservoir scenario and help specification of the reservoir surveillance options, based on a probabilistic evaluation of leakage pathway.

A1.4.2. Risk Management & data gathering

Risk management focuses on the reduction of containment risks through the generation and evaluation of mitigation options in the development plan. The existing probabilistic model is refined with improved reactive flow modelling. Usually, no additional field data is acquired in this phase of the Opportunity Realisation Process. Existing data may be re-interpreted to obtain the non-linear formation properties of existing model layers or those of additional layers to be included. In essence, the Select Phase aims at finding a storage scenario that minimises loading of critical subsurface locations.

A1.4.3. Geometrical & in-situ stress model

A Geomec model is developed or improved by incorporation of faults that have been found critical for the containment reliability assessment. Other geometrical features and stiffness and in-situ stress contrasts that influence the stress redistribution in the subsurface are incorporated, in

particular those that influence the loading of faults, top-seal and wells. Some alternative geometrical scenarios may need to be developed to assess containment reliability. This should be based on results from previous sensitivity studies.

A1.4.4. Formation property modelling

Non-linear constitutive models should be selected to simulate post-failure behaviour of top-seals, faults, overburden and the disposal reservoir as required. This may comprise selecting shear failure models, such as Mohr-Coulomb, Cam-Clay, or an-isotropic plasticity model, and tensile failure models to simulate fracturing of intact rock. Also, different thermal expansion between rock and pore fluid, as well as any non-linear relations should be considered. Associated model parameters, including those for the discrete faults, should be determined from available log data and rock mechanical testing on available core material. The linear-elastic evaluations conducted in ORP Assess should reveal where in the subsurface post-failure behaviour should be incorporated.

A1.4.5. Evaluation Options & Reservoir scenario modelling

Focus is on the evaluation of various design options, such as alternative reservoir targets of injection (and production) wells and well trajectories. Reactive flow simulations will address different dynamic pressure, temperature and CO₂ purity scenarios, and may be conducted in (a partly) coupled fashion, e.g. between fluid, heat & chemical transport. The tank model is abandoned in the geomechanical model by incorporation of pressure gradients over the reservoir volume.

Containment reliability is assessed using uncertainty ranges of model parameters determined in the Assess phase. The impact of uncertainty of the additional non-linear model parameters will also be evaluated. Results may trigger alternative well locations and trajectories or adjusted operating pressures in order improve containment reliability.

A1.4.6. Deliverables

The deliverables are integrated into the Field Development Plan (FDP) and should be reflected in the specification of

- The location of injection (and production) wells;
- Safe operating envelope of wells in terms of pressure, temperature or (CO₂) fluid composition;
- The dynamic reservoir pressure scenario;
- The location and accuracy of areal and in-well monitoring equipment
- The quality, capacity and/or size of hardware such as tubular and cement.

Furthermore, input is provided into the Well & Reservoir Management plan, in terms of data acquisition needs and frequency, follow-up interpretation and history matching and remedial actions to reduce containment risks and uncertainties in the Operate Phase. The WRM plan also specifies staff & skills requirements and associated OPEX contribution for managing containments risk during the injection and post-injection stage of the project.

Appendix 2. CORA Assessment tree description

A preliminary outline is given of the processes that could lead to loss of containment and is largely based on the Barendrecht Bow-Tie analysis [13], [14]. The processes are discussed according to the CORA assessment tree (Figure 3.5). An extensive overview of the long-term containment aspects is available online through Quintessa's Features, Events and Processes (FEP) database [15]. However, this database focuses on the long-term safety and performance of the storage system after the injection stage. Further work is required to identify the potential leakage risks to be evaluated in development projects and to specify the evaluation workflow in terms of data acquisition and geomechanical and reactive flow modelling.

A2.1. Containment risk

The TESLA hypothesis related to containment declares that:

- Injected (reactive) fluids will not leak significantly from the containment complex.

It is highlighted that containment should be safeguarded during the injection and post-injection stage of a CCS project. Containment risks are somewhat different during both stages, because of the different loading condition in the subsurface. However, the assessment tree is the same. So, depending on the project stage, the top event is:

- Significant CO₂ leakage from the hydrocarbon reservoir during injection phase
- Significant CO₂ leakage from the hydrocarbon reservoir in the post-injection phase

CO₂ leakage is directly associated with fluid flow out of a container complex. However, fluid flow may be triggered indirectly by change of formation properties caused by thermal or chemical processes or mechanical failure (Reactive flow diagram in Figure 1.2). Hydraulic, chemical, thermal and mechanical risks are highlighted for each of the leakage pathways mentioned in Figure 3.5.

A2.2. Fault leakage risks

The TESLA hypothesis related to fault leakage declares that

- Faults & Fractures will not create significant leak paths for non-reactive or reactive fluids (consider impact of dynamic pressure changes)

As a starting point, faults present in the storage complex need to be "proven seals" prior to the CO₂ (or H₂S) injection stage. This implies that faulted reservoirs should have a history of containing hydrocarbon gas on a geological time scale. Sealing faults may still contain fluids, as long as elongated expose to pressure gradients across them prove their low permeability. This is relevant for faults that are part of identified potential leakage pathways (system & components in Figure 1.1). Containment is not assured without the proven sealing capacity of the relevant faults at the start of the injection stage.

Fault leakage may be induced by natural causes, such as tectonic activities. It is obvious that this cause cannot be prevented or controlled in any way, but the likelihood can be assessed based on historical data and possibly by monitoring. Appropriate site selection is the only project decision that can influence this leakage risk.

Certain human activities at surface may also cause fault slippage and associated leakage, such as building (piling) activities and catastrophes, such as airplane crashes and various kinds of violence (war, terrorism). These risks are generally very small if a storage complex is selected at sufficient depth, say deeper than 300m.

Subsurface activities are the most important risk factor and are largely associated with hydrocarbons production, either in-field or in neighbouring fields. Depletion or inflation

pressures in the reservoir are the key factor as they may introduce pressure differentials across sealing faults. Obviously, containment may only be at risk if differentials are driving reservoir fluids out, across the fault or along the fault upward.

Fault permeability may be increased by mechanical damage. Reservoir depletion or inflation will introduce compressive and tensile stresses in the subsurface in addition to the existing in-situ stress condition. Fault slip (damage) may occur if the actual shear stress on a (part of the) fault plane exceeds the carrying capacity. Also, fault aperture may occur if the fault normal compressive stress disappears as a result of the stress redistribution, for instance at shallow locations and in extensional tectonic regimes. Fault slip may - but also may not - increase fault permeability and thereby the risk of fault leakage.

Injection or depletion pressure gradients around the wellbore may induce the same mechanical failure phenomena as induced by reservoir pressure changes. However, the stress changes and associated failure acts on a much smaller scale. Nonetheless, containment can be compromised in a catastrophic way.

Geochemical reactions over longer periods may change fault properties. This may increase fault permeability and potential leakage of reservoir fluids.

A2.3. Top-seal leakage risks

The TESLA hypothesis related to top-seal declares that

- Cap rocks will not leak significant volumes of CO₂ (consider matrix & existing faults/fractures under non-injecting conditions)

Capillary leakage occurs if the local pressure in the reservoir rock exceeds the capillary entry pressure of the local clay layers adjacent to cap rocks. The capillary entry pressure is dependent on various host formation properties and the penetrating CO₂ fluid composition. The leakage flow rate is proportional with cap-rock permeability and the actual pore pressure in the cap-rock in excess of the capillary entry pressure.

Local over-pressurisation may be induced by various mechanisms. One possibility is by de-watering of clay in the cap rock or reservoir. Dehydration may cause quartz particles or other fine clay material to detach and become suspended in the reservoir fluid. Upon migration, these particles may bridge pore throats (e.g. near the injection well) and reduce permeability and injectivity. This may result in local over-pressurisation.

The presence of highly permeable streaks or layers may also cause local over-pressurisation by the introduction of preferential flow and a non-uniform (pressure) distribution of CO₂/H₂S. Natural, existing reservoir fractures have the same effect. These features may rapidly lead to local (and temporary) over-pressurisation of the top-seal although much more storage capacity is available. Chemical reactions (CO₂ in water and reaction products with oil), enhanced by favourable pressure and temperature conditions, may introduce or increase permeability contrasts, e.g. by permeability / porosity degradation. All these effect may locally exceed the capillary entry pressure and induce fluid flow out of the reservoir.

Shear failure (damage) may occur if the actual shear stress in (a part of) the top-seal exceeds the shear stress capacity. This leakage risk is similar to that discussed for faults earlier, except that fault slippage occurs along an existing plane of weakness with a given orientation, instead of creation of a failure plane in intact rock.

Similarly, fracturing may occur if the actual tensile stress in a (part of the) top-seal exceeds the tensile strength. Both, inflation and depletion pressure changes in the reservoir may cause the compressive in-situ stress to disappear and tensile stresses to develop. Fracture propagation may

occur if the pressure in the newly created or existing (natural reservoir) fracture exceeds the "Frac propagation pressure".

Fractures have the greater likelihood of propagating around injection wells where pore pressure is highest. Fracturing may be triggered due to fluctuations in injection rate, which typically occur after starting and stopping injection in a particular well. CO₂/H₂S phase changes, caused by these fluctuations, may aggravate pressure and temperature loading. These dynamic loading conditions may trigger fractures to initiate into the top-seal.

Change of the CO₂/H₂S injection temperature may cause thermal expansion or contraction that could introduce additional stresses that contribute to fracturing or shear failure of the reservoir and top-seal. Furthermore, exothermic reactions between water and CO₂ (H₂S?) may be triggered causing additional temperature change.

Shrinkage and swelling of clay in the top-seal may induce additional stress increments and may alter various formation properties. Shrinkage may be induced by exposure to CO₂/H₂S, which cause dewatering of clay. Swelling of the clay in the cap rock may occur if water is taken along with the injected CO₂/H₂S. Differential swelling between layers may cause fracturing or shear failure due to local stress perturbations. Swelling may also lead to reduced permeability and reservoir injectivity.

A2.4. Well leakage risks

The TESLA hypothesis related to well leakage declares that

- Wells will not create significant leak paths for non-reactive or reactive fluids

Leakage risks for injection well, observation wells, or abandoned wells can be classified into tubular failure risks and cement failure risks [23].

Loss of casing and tubing integrity may be caused by corrosion, erosion, as well as mechanical and thermal loading phenomena. Long-term exposure to CO₂ and H₂S gas have a known negative effect on corrosion and brittleness of steel, which may be aggravated by temperature changes of injected CO₂ and reservoir compaction / inflation.

Chemical degradation of cement, leaching processes and mechanical failure may cause loss of zonal isolation between formation layers. Mechanical failure may be induced by change of reservoir pressure and by the pressure gradient in the near-wellbore region. Overburden and reservoir formations react mechanically to the changed fluid pressure causing a mechanical loading to the cement. Cracks in the cement, between cement and formation and between cement and tubular may create leakage pathways and loss of zonal isolation.

Appendix 3. Deterministic engine

A3.1. Stress and displacement calculation using the Geertsma model

The Geertsma model [18], [20] assumes a horizontal disc-shaped reservoir with radius R and height h at a depth D . The homogenous in-situ stress condition in reservoir and overburden is specified by the gradients for vertical and horizontal stress and the pore pressure, represented by $S_{v\text{-grad}}$ and $S_{h\text{-grad}}$ and $P_{p\text{-grad}}$. Any difference in horizontal stress cannot be accounted for due to the axi-symmetrical nature of the model. The overburden and reservoir behaviour is assumed to be homogenous and linear-elastic, specified by the same Young's modulus E and Poisson's ratio ν . The displacements and stress changes outside the disc-shaped reservoir induced by a reservoir depletion ΔP (negative for inflation) is calculated according to Geertsma's nucleus of strain concept.

An implementation of the Geertsma model has been made available in Microsoft Excel's Visual Basic for Applications (VBA). The subroutines have been collected in a separate VBA module and are described in Table A3.1. W. Pao has been instrumental in developing and debugging the subroutines for the integration of the elliptical integrals. The module can be included in any Microsoft Excel workbook that supports VBA. The input example is given in Figure A3.1.

Table A3.1: Overview of Visual Basic subroutines to calculate the displacements and incremental, total and effective stress condition in a Geertsma model of the subsurface.

	Subroutine or Function name	Purpose
1	Geertsma_DisplacementVr	Calculates the (incremental) displacements according to the Geertsma nucleus of strain model at a given evaluation point (x_r , z_r) induced by reservoir pressure change (ΔP). Returns a pointer to a 1x2 displacement (row) vector (u_r , u_z).
2	Geertsma_DispZcentre_analytical	Calculates the vertical displacement along the axis of symmetry above the disc-shaped reservoir using the Geertsma nucleus of strain model at a given depth (z) induced by reservoir pressure change (ΔP). Returns (u_z).
3	Geertsma_StressMx_increm Geertsma_StressVr_increm	Calculates the incremental stress change according to the Geertsma nucleus of strain model at a given evaluation point (x_r , z_r) induced by reservoir pressure change (ΔP). Returns 3x3 stress matrix or 1x4 row vector.
4	Geertsma_StressMx_total Geertsma_StressVr_total	Same as '_increm' subroutines, but calculate the total stress. Matrix or row vector output resp.
5	Geertsma_StressMx_effective Geertsma_StressVr_effective	Same as '_increm' subroutines, but calculate the effective stress. Matrix or row vector output resp.
6	RI1, RI2, RI3, RI4, RI6, RI7, GAMA, RUNIT, ELLF, ELLE, RF & RD	Series of subroutines that take care of the integration of the elliptical integrals that occur in the analytical expressions for displacement and incremental stress.

The stress change induced by a change of reservoir pressure ΔP is calculated in the main subroutine `Geertsma_StressMx_increm` in a point $\mathbf{P}$ with place vector $\mathbf{x}_p = (r_p, z_p)$. The result is delivered in 3x3 matrix form. The `Geertsma_StressVr_increm` provides the same results in 1x4 row format to facilitate convenient output into Microsoft Excel. Similarly, the *total stress* condition is calculated by `Geertsma_StressMx_total` from the vertical and horizontal in-situ stress gradients $S_{v\text{-grad}}$ and $S_{h\text{-grad}}$, according to

$$\sigma_{rr} = z_p \cdot S_{h\text{-grad}} + \Delta\sigma_{rr} \quad (\text{A3.1})$$

$$\sigma_{\theta\theta} = z_p \cdot S_{h\text{-grad}} + \Delta\sigma_{\theta\theta} \quad (\text{A3.2})$$

$$\sigma_{zz} = z_p \cdot S_{v\text{-grad}} + \Delta\sigma_{zz} \quad (\text{A3.3})$$

Herein, $\Delta\sigma_{rr}$, $\Delta\sigma_{\theta\theta}$, and $\Delta\sigma_{zz}$ are respectively the incremental radial, tangential and vertical stress components that are calculated by `Geertsma_StressMx_increm`.

The *effective stress* condition is calculated from the total stress components and the pore pressure gradient $P_{p\text{-grad}}$:

$$\sigma'_{ii} = \sigma_{ii} - z_p \cdot P_{p\text{-grad}} \quad (\text{A3.4})$$

with σ'_{ii} the effective stress and σ_{ii} the total stress components, with $i = r, \theta, z$.

Base Case Description				Top-seal Evaluation			
Reservoir Pressure				Top-seal location			
Reservoir Depletion	Dp		5.0 MPa	Definition point	coordinates	Radius [m]	Depth [m]
Disc Geometry				Definition point	start / center	center	
Reservoir Depth	D		2000 m	Dip with vertical	anti-clockwise	90.0 deg	
Reservoir Radius	R		2000 m	Length		500.0 m	
Reservoir Height	h		100 m	Number of segments		100 Segments	
In-Situ Stress				Shear failure properties			
Vertical Stress Gradient	SvGrad		22.5 kPa/m	Cohesion	Coh	5.00 MPa	
Horizontal Stress Gradient	ShGrad		15.8 kPa/m	Friction Angle	Phi	22.0 deg	
Pore Pressure Gradient	PpGrad		10.5 kPa/m				
Elastic Properties subsurface							
Compressibility	Cm		170.0E-6 MPa				
Poissons Ratio	Poisson		0.25 -				

Figure A3.1: Model parameters of the Geertsma's nucleus of strain model (left) and the description of the evaluation line (top-seal) and associated shear failure properties (right)

A3.2. Mohr-Coulomb shear failure criterion

The shear loading condition of a material point in the subsurface can be evaluated against various shear failure criteria, such as Mohr-Coulomb, without doing a proper nonlinear analysis. The approach outlined is equally applicable to results obtained from a Geertsma, QuickBlocks or a (linear-elastic) Geomec analysis.

The Mohr-Coulomb shear failure criterion is specified by the cohesion C and the friction angle φ , as outlined by the Mohr-circle diagram in Figure A3.2. The stress condition of a material point is represented by three Mohr circles of which the largest (between maximum and minimum principle stress components σ_1 and σ_3) is drawn in the diagram. A material point under consideration is perceived in an elastic state of deformation if the Mohr circle remains below the failure line, whereas the material is in shear failure if the circle touches the failure line. Stress conditions that cause the Mohr circle to intersect the failure line are not feasible.

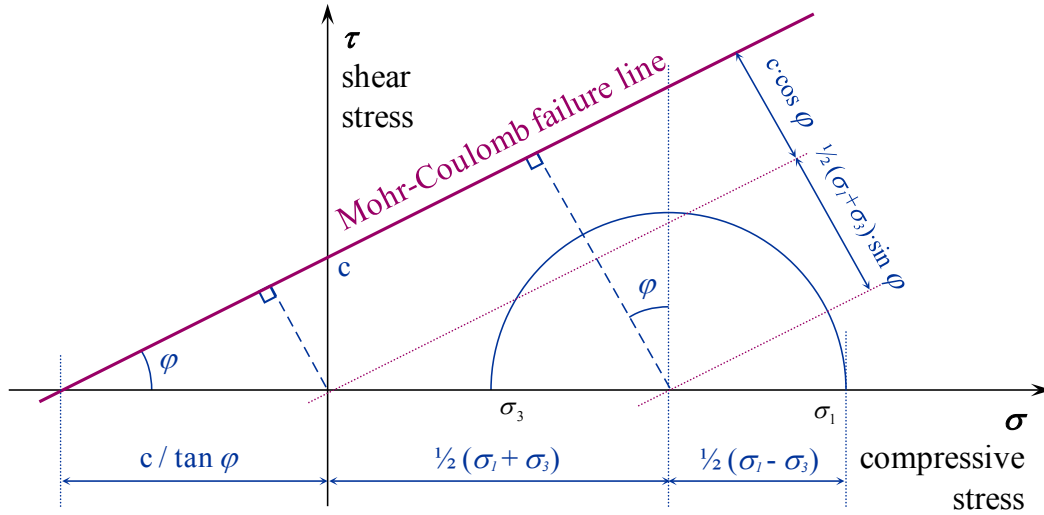


Figure A3.2: Mohr-circle for principal stress components σ_1 and σ_3 and the Mohr-Coulomb failure line diagram for cohesion C and the friction angle φ . The actual shear stress is represented by the radius of the Mohr circle, whereas the Shear Capacity is represented by the dashed line through the centre.

The Shear Capacity Utilisation (SCU) relates the actual level of shear stress to the shear capacity of the material point under consideration. The shear capacity $\tau_{\max}$ can be calculated from the cohesion C and the friction angle φ as graphically indicated in Figure A3.2:

$$\tau_{\max} = C \cos \varphi + \frac{1}{2}(\sigma_1 + \sigma_3) \sin \varphi. \quad (\text{A3.5})$$

Herein is σ_1 the largest and σ_3 the smallest (compressive) principal stress component. The actual shear stress τ is equal to the radius of the Mohr circle:

$$\tau = \frac{1}{2}(\sigma_1 - \sigma_3), \quad (\text{A3.6})$$

so that the Shear Capacity Utilisation is calculated by:

$$SCU = \frac{\tau}{\tau_{\max}} = \frac{\frac{1}{2}(\sigma_1 - \sigma_3)}{C \cos \varphi + \frac{1}{2}(\sigma_1 + \sigma_3) \sin \varphi} \quad (\text{A3.7})$$

The SCU expresses the loading condition of a material point as a fraction of its carrying capacity and has a value between 0 and 1. However, the ratio for $\tau/\tau_{\max}$ can become larger than 1 in the absence of an appropriate stress redistribution algorithm, which is the case when employing linear elasticity theory as in Geertsma's approach. In such a case, $SCU \geq 1$ is interpreted as a material point in shear failure and allows a quick screening of critical locations in the subsurface. A proper non-linear analysis should be conducted - using Geomec for example - to include post-failure behaviour in the determination of the geomechanical response.

An example of the calculation of the Shear Capacity Utilisation is provided in Figure A3.3, which shows the normal and shear distribution as a function of the along plane coordinate. The case is the same as in Figure 4.2. The maximum shear stress is found at the edge of the depletion area ($s=250\text{m}$) as expected and fades out further away. The normal stress approaches the vertical in-situ stress at $(22,5 - 10,5 \text{ kPa/m}) * 1980\text{m} = 23,76 \text{ MPa}$ for large s -coordinates. Some disturbance is found just above the edge. The SCU is calculated using eq. (A3.7) and has a maximum of 0,89 at 5 MPa reservoir depletion. The onset of shear failure at this plane is found at about 6.5MPa depletion. This can be derived from Figure A3.4, in which the response parameters are plotted as a function of the reservoir depletion pressure. The maximum SCU is larger than 1 and the failure

area starts to develop for a depletion larger than about 6.5 MPa. The failure area is about 17% or 85m at 15 MPa depletion.

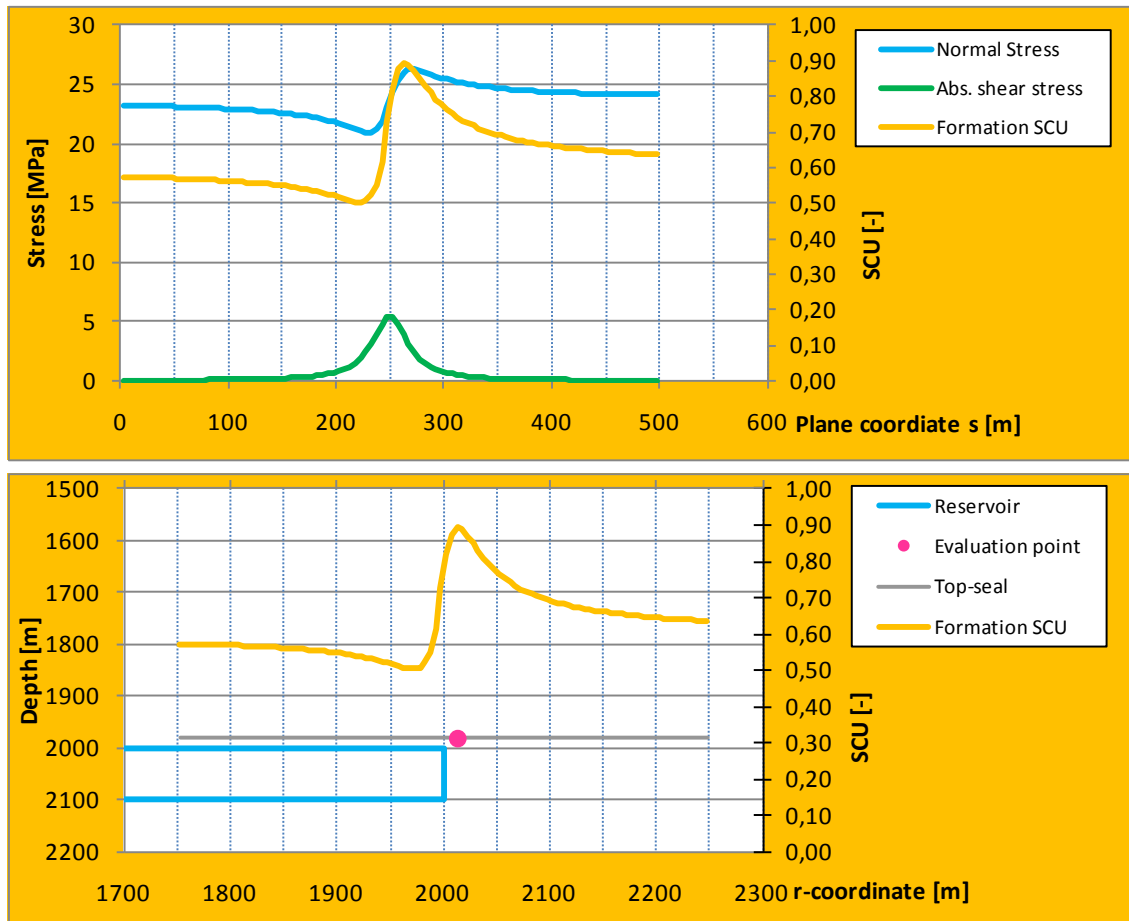


Figure A3.3: The normal and (absolute) shear stress distribution according to Geertsma's nucleus of strain model along a plane in the top-seal at 1980m depth (grey line) and at 5 MPa depletion (top) and the associated Shear Capacity Utilisation (bottom). The maximum SCU value is 0,89 (right-hand scale).

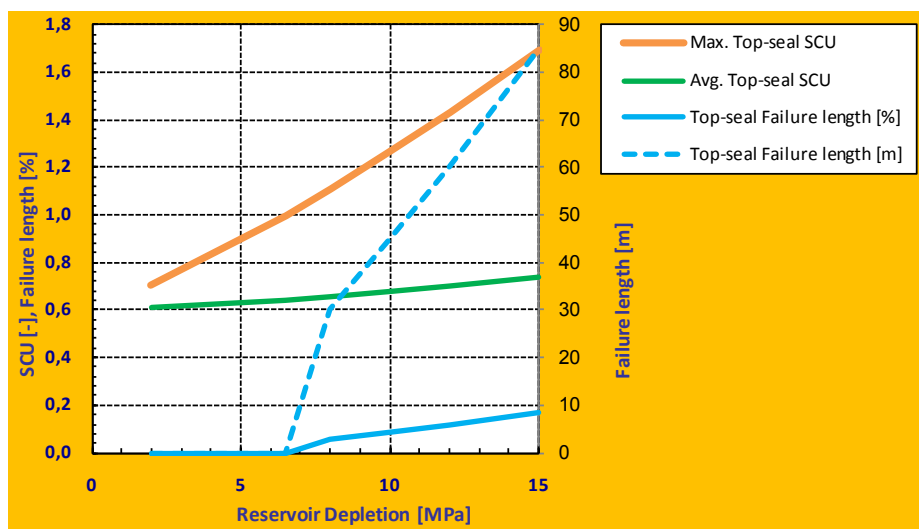


Figure A3.4: The four response parameters as a function of reservoir depletion pressure.

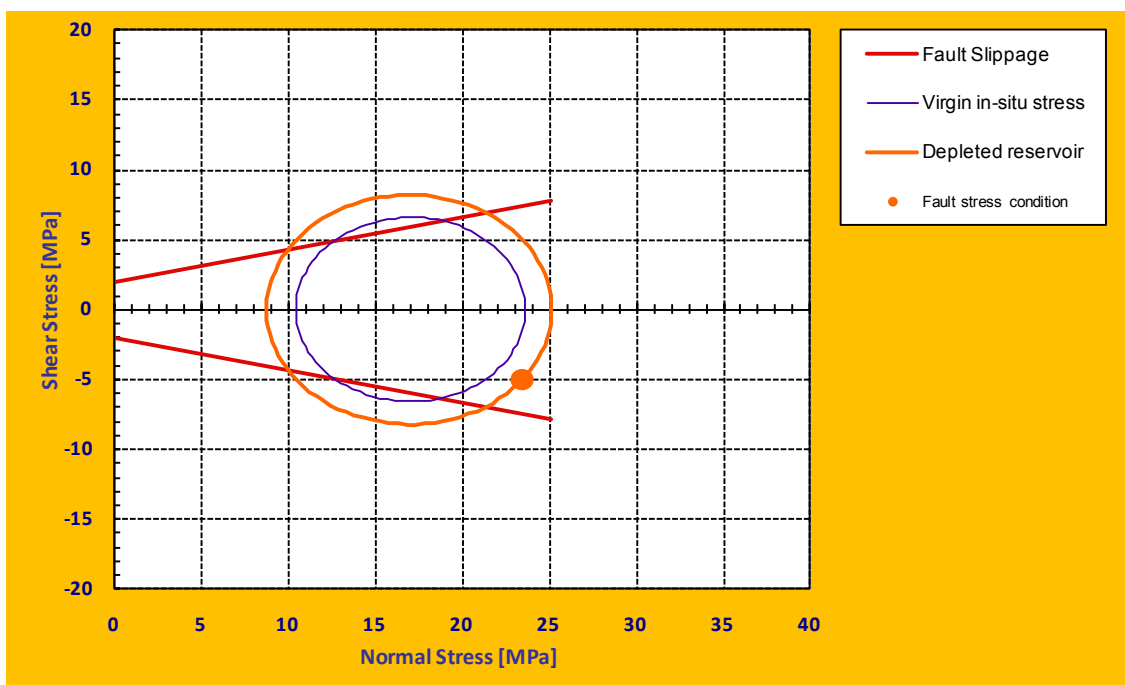


Figure A3.5: Mohr circle and stress point at the evaluation point (Figure 4.3) at 5 MPa depletion. The material point on the fault is not slipping, because the fault stress condition falls within Mohr-Coulomb failure envelope. The fault would be slipping at the same evaluation point if oriented differently.

Appendix 4. CORA Software

A4.1. Reliability Assessment of an individual leakage pathway

This Appendix describes the software structure of the lab tool for COntainment Reliability Analysis (CORA), which has been developed in Microsoft Excel Visual Basic for Applications (VBA). The lab tool incorporates the fast probabilistic geomechanics workflow developed by Pao [22] and distinguishes the following steps (Figure A4.1):

1. Specification of the reliability model
2. Generation of a Scenario Table for deterministic analysis
3. Calculation of the Scenario Results
4. Calculation of the Proxy Function
5. Reliability calculation based on the selected limit state functions

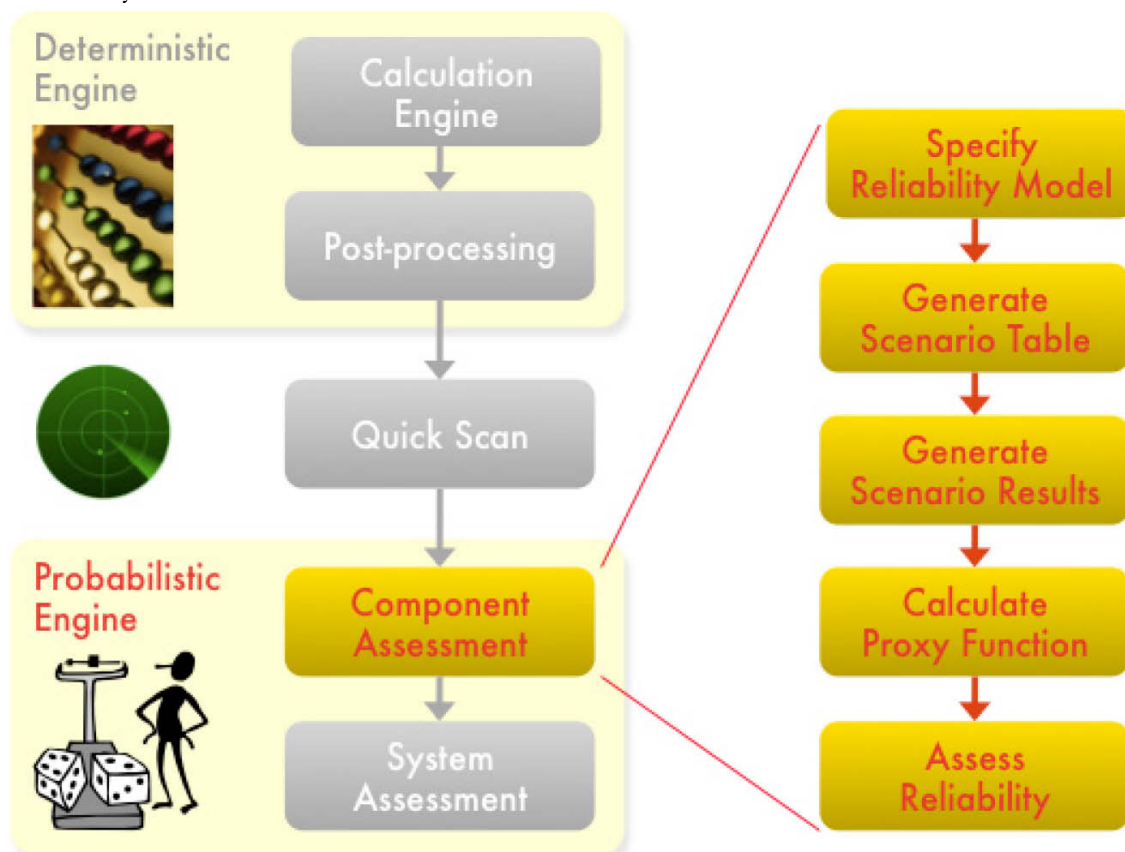


Figure A4.1: The Five building blocks of CORA's Component Reliability Assessment module

The Component Assessment module is coded in five building blocks, one for each step in the work workflow. Each building block comprise of a main program that takes care of the I/O between the user-interface (Excel) and the functional routines (Visual Basic). The separation between user interface and calculation engines simplifies re-use of the code in a different spreadsheet, if desired. Communication between building blocks is conducted through Excel by making use of so-called named ranges. The main data that is being passed through between building blocks is specified in Table A4.2. Each building block is described in more detail in the next sections, with focus on the evaluation of geomechanical aspects and fault stability in particular.

Table A4.2: Input & Output data for each of the Building Blocks. In red text indicates user input data

Building block	Input	Output
Specify Reliability model	• Input (filename) of Geertsma, QuickBlocks or Geomec model • Select wells, faults and tops-seals for assessment • Provide uncertainty range for selection of model parameters	• Model parameters for geometry, in-situ stress, formation properties and reservoir scenario • Specification of wells, faults and tops-seals to be assessed • Probability Distribution Function of selected model parameters
Generate Scenario Table	• Probability Distribution Function of selected model parameters	• Scenario Table of cases to be evaluated by specified deterministic calculation engines and associated values for uncertain model parameters.
Generate Scenario Results	• Scenario Table • Response parameters to be calculated by specified deterministic calculation engines	• Table of response parameters
Calculate Proxy Function	• Table of response parameters	• Algebraic expression as proxy for the calculation of each response parameter
Evaluate Component Reliability	• Wells, faults and tops-seals to be assessed, and • Corresponding definition of failure (Limit-state function) • Proxy-function of response parameters	• Reliability (probability of failure) for each well, fault or top-seal specified

A4.2. Specification of the Reliability Model

The reliability model specifies the deterministic model and the associated uncertainty range of model parameters. Uncertainties can apply to the geometry, the in-situ stress, the formation properties or the (dynamic) reservoir pressure. If desired, only a part of the model parameters can be identified as uncertain. Currently, only deterministic models for the Geertsma and Geomec calculation engines can be handled, but QuickBlocks and reactive flow models can be added. The uncertain model parameters in Table A4.3 correspond to the case discussed in section 4.4.

Table A4.3: Model uncertainty parameters and their probability distribution function. The P10, P50 and P90 values are redundant, but are included to reflect original input data.

Uncertainty Parameter	P50	P10	P90	Distribution	Par-1	Par-2	Par-3	Par-4
F_Coh	4	2	6	Normal	4	1,5606	0	0
F_Phi	20	15	25	Normal	20	3,9015	0	0
Dp	10	8	12	Normal	10	1,5606	0	0
Poisson	0,25	0,2	0,3	Normal	0,25	0,039	0	0
Cm	0,00017	0,00012	0,00022	Normal	0,0002	4E-05	0	0
Coh	4	2	6	Normal	4	1,5606	0	0
Phi	28	24	32	Normal	28	3,1212	0	0

The reliability model also specifies the response parameters requested from the deterministic calculation engine as given in Table A4.4. The fault name, the requested response parameter and calculation engine are given in columns 1, 2 and 3 respectively. Currently, four response parameters for faults can be calculated using the Geertsma and Geomec calculation engines as given in the table.

Table A4.4: Leakage pathways and requested response parameters to be evaluated by the specified calculation engine.

Component	ResponseParameter	Engine
Fault_1	Fault_SCU_Average	Geertsma
Fault_1	Fault_SCU_Maximum	Geertsma
Fault_1	Fault_Failure_area_m	Geertsma
Fault_1	Fault_Failure_area_pct	Geertsma

The **Fault_SCU_Maximum** response parameter provides the maximum Shear Capacity Utilisation (SCU) on a fault plane. The SCU is defined as the actual effective shear stress divided by the maximum allowable shear stress in a material point:

$$SCU = \frac{\tau}{\tau_{\max}} \quad (\text{A4.1})$$

A material point is at the onset of shear failure if $SCU=1$, whereas a value smaller than 1 indicates intact material with a linear-elastic response. Values larger than 1 can be calculated if linear-elastic formation behaviour is assumed. Selection of an appropriate plasticity (shear failure) model takes care of redistribution of shear stress loading such that for no material point $SCU>1$. A non-linear analysis may be warranted if **Fault_SCU_Maximum**>1 is found. The maximum shear stress or shear stress capacity can be calculated by various criteria, but currently only Mohr-Coulomb is provided.

The value for **Fault_SCU_Maximum** provides information of the most critically loaded point on a fault plane, and the **Fault_SCU_Average** about the average SCU over the entire fault plane. The **Fault_Failure_area_m** and **Fault_Failure_area_pct** provide information about the area of the fault plane that is in failure, respectively in m² or as percentage of the entire fault plane. For a linear-elastic model this is equivalent with the areas for which $SCU \geq 1$, for a plasticity model this is for $SCU=1$.

The reliability model also defines operational failure for each well, fault, or top-seal. Operational failure is assessed by a limit-state function LSF , which evaluates a solicitation value S against a allowable resistance R :

$$LSF = R - S. \quad (A4.2)$$

Operational failure occurs if $LSF < 0$. The solicitation value, which is a function of response parameters, and the resistivity have a probability distribution function.

It is noted that operational failure is not the same as material failure. Material failure is embedded in the response function R and is determined by the specified formation properties. A fault may or may not exhibit material failure before operational failure is declared, depending on the limit value L . The limit-state function that specifies operational failure should incorporate any regulatory requirements.

In summary, a reliability model specifies

- the deterministic model and associated calculation engine,
- the uncertainty ranges of a selection of model parameters,
- the response parameters and associated material failure properties to be evaluated for each selected of fault, well or top-seal,
- a limit-state function that defines operational failure

Table A4.5: VBA subroutines for Generating the Reliability Model.

	Subroutine or Function name	Purpose
1	GeertsmaModel_ UncertaintyParameters_write	Writes table of uncertainty parameters to Excel sheet (Table A4.3) from Geertsma model description (Figure 4.4)
2	UncertaintyParameters_select	Selects uncertain model parameters from generic Excel list

A4.3. Generating the Scenario Table

A scenario table describes a set of cases studies to be evaluated using the specified deterministic calculation engine. A scenario or case study is specified by the values of the uncertain parameters in the model, such as Young's modulus, Mohr-Coulomb friction angle, or minimum horizontal in-situ stress (see Figure A4.1). The number of uncertain parameters can vary, but is limited to 20 in the current implementation. Geometrical uncertainties, such as reservoir thickness or fault dip angle, may also be taken as an uncertain parameter using the Geertsma engine. However, dedicated geometrical realisations are required for most other calculation engines.

Table A4.6: Scenario Table for the Geertsma model in Figure 4.4.

Case	F_Coh	F_Phi	Dp	Poisson	Cm	Coh	Phi
1	2	15	12	0,3	0,00022	2	32
2	6	25	12	0,2	0,00022	6	24
3	4	20	10	0,25	0,00017	4	28
4	2	25	12	0,2	0,00022	2	24
5	4	20	10	0,25	0,00017	4	28
6	2	15	8	0,2	0,00012	2	24
7	6	15	8	0,2	0,00022	6	32
8	6	25	8	0,3	0,00012	2	24
9	6	15	12	0,3	0,00012	6	24
10	6	25	8	0,3	0,00022	2	32
11	2	15	8	0,3	0,00022	6	24
12	2	25	8	0,2	0,00012	6	32
13	6	15	12	0,2	0,00012	2	32
14	2	25	12	0,3	0,00012	6	32

The scenario table in Table A4.6 is generated by the “Generate Scenario Table” building block. Each row describes a case study. The first column contains the case study number, whereas the other columns contain the values of the uncertain model parameters. Base case values are taken for all model parameters not specified in the scenario table. The first row (header) of the scenario table specifies the labels of the uncertain model parameters.

The scenario table is generated using template tables obtained from experimental design techniques [21]. A template table describes a complete set of scenarios for any given number of uncertain model parameters (Table A4.7). The template specifies if the P10 value, the mean value or the P90 value of an uncertainty parameter should be taken in a particular scenario. The number of scenarios in the template is determined by experimental design techniques and is entirely based on the number of uncertain parameters. The selected template table is therefore independent on the type of engine or model that is being employed.

Table A4.7: Template Experimental Design Table for 7 uncertain model parameters. The experimental design method prescribes 14 “experiments” to be conducted with the specified values for each uncertain parameter. Key: “-1” represents the P10 value, “0” represents the P50 value and “1” represents the P90 value.

<i>Exp #</i>	<i>F1</i>	<i>F2</i>	<i>F3</i>	<i>F4</i>	<i>F5</i>	<i>F6</i>	<i>F7</i>
1	-1	-1	1	1	1	-1	1
2	1	1	1	-1	1	1	-1
3	0	0	0	0	0	0	0
4	-1	1	1	-1	1	-1	-1
5	0	0	0	0	0	0	0
6	-1	-1	-1	-1	-1	-1	-1
7	1	-1	-1	-1	1	1	1
8	1	1	-1	1	-1	-1	-1
9	1	-1	1	1	-1	1	-1
10	1	1	-1	1	1	-1	1
11	-1	-1	-1	1	1	1	-1
12	-1	1	-1	-1	-1	1	1
13	1	-1	1	-1	-1	-1	1
14	-1	1	1	1	-1	1	1

Table A4.8: VBA subroutines for Generating the Scenario Table.

	Subroutine or Function name	Purpose
1	ScenarioTable_Generate_start	Provide a VBA entry point for Excel sheet button; collects all relevant input data and launch <ScenarioTable_Generate>; Write results to Excel (Table A4.6)
2	ScenarioTable_Generate	Generate Scenario Table from template by substitution with real parameter values.
3	DesignScheme_Select	Selects the appropriate Template Experimental Design Table (Table A4.7) depending on the number of uncertain model parameters from available collection (up to 20 uncertain parameters)

A4.4. Calculating the Scenario Results

Response parameters are calculated for each of the case studies specified in the scenario table. A response parameter could be the maximum Shear Capacity Utilisation SCU on a fault plane, the smallest (most tensile) stress in the top-seal, the largest compaction strain along a well trajectory, or any other result that can be derived from a geomechanical or non-geomechanical calculation engine. The response parameters are used to evaluate one or more limit state functions (see next section) and are collected in a Scenario Results Table (Table A4.9). The associated labels are stored in the first row of the scenario results table.

Table A4.9: Scenario Results associated with the scenario table in Table A4.6.

Engine Component	Geertsma Fault_1	Geertsma Fault_1	Geertsma Fault_1	Geertsma Fault_1
Case	Fault_SCU_Average	Fault_SCU_Maximum	Fault_Failure_area_m	Fault_Failure_area_pct
1	0,91385263	2,01154573	50	0,25
2	0,4320081	0,95811757	0	0
3	0,5039167	0,98208101	0	0
4	0,60166501	1,3157842	12	0,06
5	0,5039167	0,98208101	0	0
6	0,67534764	1,10677296	6	0,03
7	0,4915207	0,97606144	0	0
8	0,3163038	0,52063954	0	0
9	0,46121681	0,86166928	0	0
10	0,37260907	0,73512732	0	0
11	0,79206063	1,55321521	28	0,14
12	0,44771394	0,73109343	0	0
13	0,46131701	0,86177087	0	0
14	0,49354703	0,90764794	0	0

The limit state functions selected in the input determine which response parameters are to be calculated. Multiple limit-state functions may be selected to evaluate the containment reliability of faults, top-seals or wells. That is, multiple wells, faults or top-seals may be selected for evaluation of e.g. the maximum SCU value. This implies that a deterministic calculation engine, such as Geomec, can deliver various response parameters for multiple components in a single deterministic analysis.

In principle, more than one deterministic calculation engine can be invoked to produce the requested response parameters. However this capability is not available in the current lab tool. This would allow comparison of the performance of alternative engines, such as Geertsma, QuickBlocks and Geomec, in calculating component reliability in relation to their modelling assumptions. Additionally, non-geomechanical engines, such as a reservoir simulator, a geochemical or well integrity engine can be invoked to produce different kinds of response parameters. The user interface should facilitate the selection of response parameters and associated calculation engines.

The software design is flexible to incorporate new response parameters and associated calculation engines as required. This implies that additional pre-processing can be added to launch new calculation engines and that dedicated post-processing can be added to calculate newly defined response parameters. Obviously, this also includes the possibility to define new response parameters using existing calculation engines.

Table A4.10: VBA subroutines for Calculating the Scenario Results.

	Subroutine or Function name	Purpose
1	ScenarioResults_Generate_start	Provide a VBA entry point for Excel sheet button; collects all relevant input data and launch <ScenarioResults_Generate>; Write results to Excel (Table A4.9)
2	ScenarioResults_Generate	Compile list of calculation engines to be invoked; Collect response parameters from each engine in table
3	ResponseParameters_Geertsma_Calculate	Calculate the specified Response Parameters for each scenario and specified faults, wells and top-seals according to the <u>Geertsma</u> 's nucleus of strain model.
4	GeertsmaModel_initiate	Initiates a Geertsma model with scenario values
5	GeertsmaModel_Fault_read	Defines fault plane (cone) in Geertsma model by its end point coordinates, number of segments and Mohr-Coulomb properties
6	GeertsmaModel_Fault_Stress_Calculate	Calculates the stress components in the fault segment coordinates according to the Geertsma's nucleus of strain model.
7	ResponseParameters_QuickBlocks_Calculate	Calculate the specified Response Parameters for each scenario and specified faults, wells and top-seals using <u>QuickBlocks</u> . (not available yet)
8	ResponseParameters_Geomec_Calculate	Calculate the specified Response Parameters for each scenario and specified faults, wells and top-seals using <u>Geomec</u> . (not available yet)
9	Engine_select	Compile list of engines to be invoked
10	Fault_FailureCondition	Calculates the fault Response Parameters in X based on the stress components in all segments

A4.5. Calculating the Proxy Function

A Proxy function is a substitute for the deterministic calculation engine to calculate a particular response parameter [22]. A proxy function is an algebraic expression that relates the uncertain model parameters to one response parameter R . Here, a linear series expansion is used for the proxy functions:

$$R = c_0 + \sum_{i=1}^n c_i \cdot p_i, \quad (\text{A4.3})$$

Herein, c_i are constants that are calculated using a multi-variable regression method, and p_i are the model parameters for which an uncertainty range has been specified.

Table A4.11: Proxy Function coefficients for each response parameter (columns).

Engine Component	Geertsma Fault_1	Geertsma Fault_1	Geertsma Fault_1	Geertsma Fault_1
Coefficient	Fault_SCU_Average	Fault_SCU_Maximum	Fault_Failure_area_m	Fault_Failure_area_pct
Constant	0,81270552	0,839301808	-9,276190476	-0,046380952
F_Coh	-0,0578838	-0,11302806	-4	-0,02
F_Phi	-0,0188578	-0,03671042	-1,2	-0,006
Dp	0,01116878	0,05390107	1,166666667	0,005833333
Poisson	0,40002927	1,067074243	100	0,5
Cm	1247,11652	4267,095729	140000	700
Coh	-0,0092928	-0,02349316	-1,666666667	-0,008333333
Phi	-0,0020425	-0,0019365	0,083333333	0,000416667

Table A4.12: Scenario Results obtained using the Proxy Function for each response parameter.

Engine Component	Geertsma Fault_1	Geertsma Fault_1	Geertsma Fault_1	Geertsma Fault_1
Case	Fault_SCU_Average	Fault_SCU_Maximum	Fault_Failure_area_m	Fault_Failure_area_pct
1	0,85852388	1,85933116	38,85714286	0,194285714
2	0,37757659	0,854926625	-6,476190476	-0,032380952
3	0,53335684	1,035971966	6,857142857	0,034285714
4	0,64628314	1,401011493	16,19047619	0,080952381
5	0,53335684	1,035971966	6,857142857	0,034285714
6	0,66547443	1,125801887	9,523809524	0,047619048
7	0,50513926	0,990934588	1,523809524	0,007619048
8	0,28536405	0,413292823	-8,476190476	-0,042380952
9	0,48144594	0,902028724	1,523809524	0,007619048
10	0,39373544	0,824510392	6,19047619	0,030952381
11	0,79301769	1,565246257	26,85714286	0,134285714
12	0,42338476	0,649233008	-8,476190476	-0,042380952
13	0,46227407	0,873801922	-1,142857143	-0,005714286
14	0,50806283	0,971544713	6,19047619	0,030952381

The performance of the proxy function to replace the calculation engine is assessed by comparison with the deterministic scenario results (Figure A4.2). The results are shown graphically for each response parameter. The proxy-function constants in Table A4.11 have been used to calculate scenario results in Table A4.12. It is seen that the failure area is less appropriate to capture by an algebraic expression because of the discontinues nature of the response parameter: the failure area is negligible if the maximum Shear Capacity Utilisation remains smaller

than 1. The associated proxy functions, therefore, are over-estimating the failed area at low values, and under estimating at larger values. This seems not the case for the continuous response function for the maximum and average SCU on the fault plane. Further investigation is required to assess the robustness of proxy function to replace calculation engines for response parameters for faults, wells and top-seals.

Calculation time of the response parameters of the 14 scenarios using the Geertsma model takes about 7 sec, whereas the results are obtained instantaneous using the proxy function. Time savings are much more significant in case more sophisticated Geomec or QuickBlock models are used.

The impact of each uncertain model parameter on a response parameter is displayed in a so-called Tornado chart (Figure 4.5). The coefficients in the proxy function (Table A4.11) represent the sensitivity of the response value to each model parameters. Substitution of the average values for the model parameters yields the expected (average or P50) value of the response value. The upper and lower bound values of the response parameter is obtained by substitution of the P10 and P90 values for one model parameter, while keeping other model parameters at the P50 value.

Table A4.13: VBA subroutines for Calculating the Proxy Function.

	Subroutine or Function name	Purpose
1	ProxyResults_Generate_start	Provide a VBA entry point for Excel sheet button; collects all relevant input data and launch <ProxyResults_Generate>; Write results to Excel (Table A4.9)
2	ProxyResults_Generate	Calculate the Response Parameters based on proxy functions
3	ProxyResults_NormDist	Returns the expectation value and standard deviation of a given proxy function.
4	ProxyCoefficients_Calculate	Calculate the proxy function coefficients c_i in eq. (A4.3) for each Response Parameter, based on the Scenario Results and using Multi Variable Regression
5	ProxyFunction_value	Calculates the value R of the proxy function according to eq. (A4.3).
6	TornadoChart_series	Calculates the value range of each uncertain model parameter to plot a Tornado Chart (Figure 4.5)

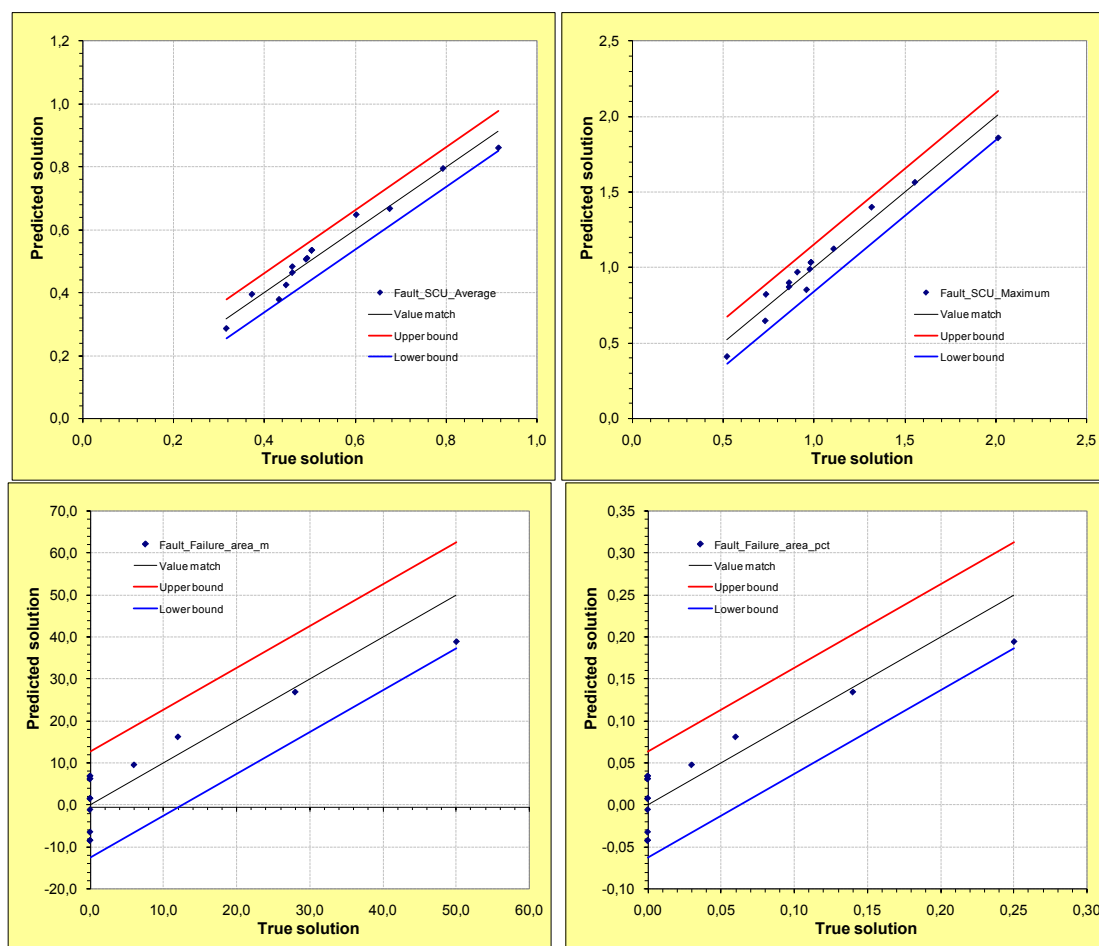


Figure A4.2: Comparison between the Fault's maximum and average Shear Capacity Utilisation (SCU) at the top, and the Fault failure area in [m] and [%] (bottom), obtained from the Proxy Functions and by the Geertsma calculation engine (Table A4.9 and Table A4.12). The upper and lower bound indicates an error of twice the standard deviation.

A4.6. Evaluating Component Reliability

The reliability of a component, that is a fault, well or top-seal, is calculated using the proxy-function. Probability of failure (reliability) is evaluated using a limit-state function that represents the definition of operational failure, taking into account the probability distribution function of the uncertain model parameters.

Each row in Table A4.14 specifies the distribution function for a particular uncertainty parameter and the associated proxy-function constant. The last two columns specify the so-called design point, where the reliability index is being calculated [22]. An iterative approach is followed to find the point with lowest reliability. The built-in Solver function in Microsoft Excel is used for this purpose.

The reliability calculation engine is invoked several times for a range of reservoir depletion pressures and values for the limit-state function (in the case of Table A4.15). In principle, any of the uncertain model parameters can be selected. The given values in Table A4.15 are substituted in the reliability calculation engine and the Excel solver is invoked. Graphical results are shown in Figure 4.7 and Figure 4.6 for the case discussed in section 4.4.

Table A4.14: Reliability calculation engine. Input is the set of uncertainty parameters and their (normal) probability distribution. The Proxy Coefficients are taken for the selected parameter. The Excel solver find vector x-i, which minimises the reliability (here 5.448) while the limit-state function is zero.

FORM Engine

Response Parameter item	1
LimitState_value (default)	2,00
Reponse Parameter selected	Fault_SCU_Average
Engine used	Fault_1
Expectation / Std. Dev	Geertsma
Limit-state value / Function	1,000,00
Reliability / Probability of Failure	5,4482,5412E-08

8 Proxy Coefficients

12 Data columns

Uncertainty Parameter	P50	P10	P90	Distribution	Par-1	Par-2	Proxy Coefficients	Given value	n-i	x-i
Constant							0,63200618		0	1
F_Coh	4	2	6	Normal	4	1,56060829	-0,0379723	-3,335460623	-1,20535	
F_Phi	20	15	25	Normal	20	3,90152073	-0,0164307	-3,608156323	5,922703	
Dp	10	8	12	Normal	10	1,56060829	0,00949844	15	4,35764E-08	15
Poisson	0,25	0,2	0,3	Normal	0,25	0,03901521	0,29501621	0,647849828	0,275276	
Cm	0,00017	0,00012	0,00022	Normal	0,00017	3,9015E-05	989,371106	2,172639853	0,000255	
Coh	4	2	6	Normal	4	1,56060829	-0,006722	-0,590459305	3,078524	
Phi	28	24	32	Normal	28	3,12121658	-0,0013015	-0,228646233	27,28635	

Table A4.15: Table of reliability values for the case in T. The reliability is calculated for various values of the limit state value (rows) and reservoir depletion pressure (columns). The user can specify any uncertain model parameter and associated value in the table.

3 LimitState_value 7 Dp							
Reliability Results	1	5	7	9	11	13	15
0,6	2,689905933	2,29739743	2,10114834	1,90489926	1,7086605	1,5124	1,31616233
0,8	4,756026085	4,36352764	4,16727851	3,9710191	3,77477001	3,57853	3,38228216
1	6,822155924	6,42963709	6,23339833	6,03714925	5,84088983	5,64464	5,44840199

Table A4.16: VBA subroutines for calculating component reliability.

	Subroutine or Function name	Purpose
1	FORM_evaluate_start	Provide a VBA entry point for Excel sheet button; Handles all i/o with Excel, invokes routines 2 and 3 of this table.
2	FORM_engine_initiate	Initiates the FORM engine (Table A4.14)
3	FORM_evaluate	Produces the reliability values in Table A4.15.
4	FORM_solve	Invokes the build-in Microsoft Excel solver for a calculate the reliability for a particular scenario.

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